Design and Implementation of a Decentralised Solar-Based Smart Grid with Optimised Load Management

Final Year Design Project Report

(EE491 & EE492)

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Certificate of Ownership

It is certified that the work presented in this report, regarding our senior year design project titled ***"Design and Implementation of a Decentralised Solar-Based Smart Grid with Optimised Load Management"*** has been carried out through the collective efforts of all the undersigned team members. This work is carried out under the supervision of the undersigned faculty members, using number of available resources including research articles, books, and expert opinion.

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Final Year Design Project as a Complex Engineering Problem

It is to certify here that the final year design project (FYDP) entitled ***"Design and Implementation of a Decentralised Solar-Based Smart Grid with Optimised Load Management"*** is categorized as a complex engineering problem (CEP) based on the preamble (in-depth engineering knowledge) and involvement of the following attributes.

- Depth of knowledge required
- Depth of analysis required
- Familiarity of issues
- Interdependence

The above listed attributes are thoroughly assessed after conducing meeting on February 25, 2025, with the following final year students, who proposed the idea of this FYDP.

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This project is going to be conducted in semesters Fall 2025 and Spring 2026. Further, it is submitted that the proposed idea is worthy, and the required efforts are up to the level of a final year design project.

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Nomenclature

FYDP	Final Year Design Project
DCNG	DC Nano Grid
DCMG	DC Micro Grid
BESS	Battery Energy Storage System
SoC	State of Charge
MPPT	Maximum Power Point Tracking
P&O	Perturb & Observe
BDC	Bi-directional Converter
CEMS	Cloud Energy Management System
ILP	Integer Linear Programming
SDG	Sustainable Development Goals
PV	Photovoltaic
IoT	Internet of Things

Abstract

This project explores the development of a swarm-oriented DC Nano Grid (DCNG) designed to provide reliable and sustainable energy for rural and off-grid regions such as Topi, Pakistan. The proposed system adopts a decentralized approach in which each household operates its own photovoltaic (PV) generation unit paired with battery storage, enabling both self-sufficiency and peer-to-peer energy exchange within the network. To ensure efficient energy harvesting, a buck converter with Maximum Power Point Tracking (MPPT) is utilized, while voltage regulation and system stability are maintained through a Battery Energy Storage System (BESS) integrated with a bidirectional buck–boost converter and PI-based control strategy.

The architecture is further enhanced by interconnecting multiple nano grids through a shared DC bus using bidirectional converter, allowing dynamic power flow based on real-time load and generation conditions. System validation is carried out using MATLAB/Simulink simulations under different irradiance and load variations, complemented by the development of a hardware prototype involving power converter design, control implementation, and PCB fabrication. The findings demonstrate that the proposed decentralized DCNG framework is a scalable, resilient, and cost-effective solution for rural electrification.

Furthermore, the proposed system is enhanced through the integration of an Internet of Things (IoT)-enabled monitoring and control framework. A dual-microcontroller architecture, consisting of an Arduino Nano for real-time sensing and an ESP32 for communication and control, is employed to ensure efficient system operation. A cloud-based Cloud Energy Management System (CEMS) utilizing Integer Linear Programming (ILP) is implemented to optimize load management and energy sharing decisions dynamically. Real-time data acquisition, remote monitoring via a web-based dashboard, and intelligent control significantly improve system reliability, scalability, and overall energy utilization.

1 Introduction

The current chapter serves to discuss the introduction to the research project, covering the definition of the problem, the reasons for choosing this topic, as well as scope and expected results. Additionally, this chapter focuses on the relevance of the research and its application in relation to sustainable development goals. It is also discussed how complicated the problems of engineering are when discussing this project [1–5].

1.1 Background and Motivation

However, access to dependable electricity continues to pose challenges in rural parts of Pakistan, such as Topi in Khyber Pakhtunkhwa, where power cuts of up to 10–14 hours a day are typical. Education, communication, and small business activities may all be hampered by power outages, while extending the grid to these areas remains prohibitively expensive. While solar energy is plentiful in this region, standalone systems may cause inefficient energy usage because some homes produce excess energy while others do not get enough power. Hence, a more efficient energy distribution and exchange approach is required in this scenario.

In response, we propose the deployment of a Swarm DC Nano Grid (DCNG). In this approach, each house is fitted with a set of PV cells, batteries, and converters (MPPT and bidirectional buck-boost converters) linked to a DC bus. Through peer-to-peer energy sharing, surplus energy can be distributed according to demand. Moreover, P&O-based MPPT and optimized load management ensure optimal energy harvesting and consumption within this architecture. Simulations using MATLAB/Simulink and experimental results demonstrate its stable performance under various operating conditions. This approach provides a scalable, reliable, and cost-effective solution for rural electrification, promoting sustainable energy use and community-level energy independence.

1.2 Problem Statement

Traditional nano-grid technologies suffer from various shortcomings, which impair their efficiency in rural electrification projects. One major drawback is that traditional nano-grids do not have an intelligent control system, leading to a lack of real-time monitoring, efficient load balancing, and proper fault detection. This makes the optimization of the grid under different working conditions impossible. Moreover, current designs fail to adapt to changes in the load profile and renewable generation sources. The outcome is unstable power flow, inefficiency, and unreliability of the entire supply chain.

Inefficiencies in battery usage arise due to inadequate SoC management, which impairs peak power requirement fulfillment. Moreover, the traditional way of distributing energy has a problem of imbalance in the use of energy and poor utilization of solar energy that is stored in batteries. The occurrence of frequent power cuts and the absence of effective coordination in energy sharing worsen the performance of decentralization. There is a necessity for having an intelligent and collaborative method of managing energy in the decentralized DC nano-grid systems.

1.3 Scope of Work and Expected Outcome

The current project is related to the modelling and implementation of a Swarm-based DC Nano Grid (DCNG) for decentralized rural electrification with an example of Topi, Pakistan. Photovoltaic (PV) power generation is modelled, and the energy storage systems as well as energy sharing techniques for households are included. The Maximum Power Point Tracking technique based on Perturb and Observe approach is proposed to extract maximum solar power in variable environmental situations.

The system will be simulated in MATLAB/Simulink software to estimate its performance under changing irradiance and loads. Moreover, the laboratory experiment will be performed by means of PV emulators, energy storages, DC-DC converters and control circuits, and simple decision technique depending on SoC and load will be developed.

Expected results will demonstrate that a Swarm DC Nano Grid is able to provide stable voltage regulation and efficient power sharing in changing environment. Despite the laboratory scale, it will confirm the efficiency of decentralized DC grid for rural electrification purposes.

1.4 Project Contribution by group members

Table 1.1 Project Contributions

Sr. No.	**Tasks**	**Sub-Tasks**	**Assigned to**
1	Simulations	MATLAB/Simulink	Haroon Abdullah
2	Optimization & Web Dashboard	Optimization Logic Dashboard Implementation	Abdul Wadood Khan
3	PCB Designing	Designing on EasyEDA Veroboard to PCB conversion of circuits	Abdul Hadi
		Verification	Muhammad Adnan Barki
4	Circuit Design	Designing & Integration of a Sensor circuit Power Calculations	Muhammad Adnan Barki

5	Hardware	Troubleshooting & Testing	All members
		Research & Development	Abdul Wadood Khan Haroon Abdullah
		Refinement of circuit	Abdul Wadood Khan Muhammad Adnan Barki
6	Documentation	FYDP Poster FYDP Report	Haroon Abdullah

2 Literature Review

2.1 Introduction

The topic of distributed energy systems has attracted considerable scientific interest because of the ongoing shift towards renewable energy systems, specifically for remote areas. In this context, remote areas are unable to utilize centralized systems owing to high levels of power loss, costly installation, and poor performance. Hence, there has been a shift from the traditional power systems towards the use of microgrids and nano grids. This is because they can be effectively implemented in DC power architecture.

2.2 Rural Electrification: Challenges and Opportunities

The issue of rural electrification still poses a serious problem in many developing nations. Despite most people being in the rural areas, there is poor provision of adequate supply of electricity. Power cuts of up to 10-14 hours occur frequently in places like Topi during the summer seasons, thus having negative impacts on various aspects, including learning institutions, hospitals, and even the economic activities. Besides, there is the issue of cost in expanding the traditional grid network to reach such areas at an estimated cost of $2,000-$10,000 per kilometer.

Nevertheless, such areas as Topi present good opportunities for using solar energy since there is a high amount of solar radiation that averages 5.3 kWh/m²/day. Therefore, PV technology can be used for electrifying such remote areas. The main concern here is how to efficiently use renewable energy sources in such locations.

2.3 Micro Grids and Nano Grids: An Overview

Microgrids are local energy systems that comprise an energy generation and distribution facility in one place, hence making it possible for these grids to be partially or entirely independent from the main electrical supply networks. In essence, DC nano grids can be described as an improved form of microgrids, working at lower DC voltages (less than 120 volts).

The use of DC systems is advantageous in rural areas due to the nature of most renewable energy sources such as photovoltaic systems and most electrical appliances that are currently available. DC nano grids have been shown to increase the efficiency of systems through the elimination of additional steps involved in changing currents and voltage levels, which results in 10%–15% improvement in efficiencies. According to Kumar et al. (2021), there has been up to 30% decrease in cost when DC nano grids were used. Similar research done by Singh et al. (2020).

2.4 Swarm DC Nano Grids and Energy Sharing Models

The idea of Swarm DC Nano Grids builds on the conventional nano grids architecture by providing a smart connection scheme among different autonomous nodes. In this design, each house becomes an individual unit of energy production and consumption using both its storage and generation systems while simultaneously working together to create an energy sharing system. Swarm intelligence guides the behaviour of the system, which relies on agents making their choices to improve the entire system's efficiency.

The sharing of energy in this system may be achieved using power electronics technology including DC-DC converters and switching units. Various studies by Ali et al., (2022) and Habib et al., (2021) have suggested decentralized energy management schemes that employ variables like the State of Charge (SoC), real-time load requirement, and solar radiations to improve power exchange decision-making processes. Their simulation results showed load satisfaction rates above 90 percent.

A significant disadvantage in current literature is the absence of effective communication and power distribution mechanisms to enable energy exchange in low-cost rural areas. This paper intends to introduce a simpler control method using relays and SoC thresholding to allow practical energy exchange decisions in small clusters of communities.

2.5 MPPT Techniques in Rural Solar Systems

Maximum Power Point Tracking (MPPT) methods play an important role in optimizing power generation in Photovoltaic systems (PVs). MPPT allows the operating point of the PV module to be adjusted to obtain optimum power output under various environmental conditions.

Several MPPT methods can be found, but one of the most popular is Perturb and Observe (P&O). This algorithm can be easily implemented into the MPPT method and guarantees accurate results in cases of gradually changing environmental factors. However, the P&O algorithm shows slight oscillations in the operation process and decreased precision in cases of fast changes in irradiation level, which makes it suitable only for rural areas characterized by stable environments.

2.6 Hardware Implementation Trends

Recently, hardware prototypes have been investigated to verify nano grid architectures. Ramana et al, designed a DC nano grid (DCNG) prototype by connecting three houses with one central battery, obtaining 12% higher efficiency than AC systems while being not scalable and having not any independent energy management. The most recent studies seem to be focused on decentralized nano grid designs with decentralized storage and bidirectional converters, to better share energy among nodes. Building on this trend, we present a Swarm DCNG architecture and an initial hardware prototype made up of solar cells, a battery pack, DC-DC converters and a switching control module.

2.7 Gaps in Literature and Project Contribution

Even though there has been a lot of work done about solar home systems and microgrids that use AC energy, there has been little focus on decentralized DCNG that can share energy between peers. In most cases, the existing studies are under ideal conditions, without incorporation of environmental data, and without hardware level testing and validation. In addition, very few researchers have incorporated the socio-economic behavior of users in rural regions like Pakistan who tend to be unpredictable.

The project will address the challenges by doing the following:

- Generation of a simulation framework by utilizing actual values of irradiance and temperature in the Topi region.
- Synthesis and implementation of a distributed architecture to facilitate energy sharing in DCNG.
- Evaluation of the designed system using simulations as well as hardware implementation.
- Formulation of a cost-effective and scalable approach for managing power between houses.

2.8 Relation to Sustainability and SDGs

The Swarm DC Nano Grid system envisioned by this project is a technical intervention to the issue of rural energy deficiency it also is very cognizant of the high level goals for global sustainability outlined by the UN SDGs. The literature survey carried out in the earlier sections demonstrates that affordable and intelligent energy infrastructure is rare and exceptionally deficient for communities spread across off-grid areas in developing countries such as Topi in Khyber Pakhtunkhwa. Implementing by means of a decentralized system comprising photovoltaic generation, battery energy storage and IoT enabled load monitoring and management through cloud computing systems places the focus of this entire project on energy access, suitable technology deployment and community durability- the three enunciating features of the SDGs. As required by the NCRC, SDG-9 (industry innovation infrastructure) is the baseline cross cutting SDG for all Senior Year Design Projects in Electrical Engineering and in addition, the descriptive and prescriptive scope of the entire project relates it also to SDG-7 (affordable and clean energy) and SDG-11 (sustainable cities and communities). The following subsection delineates and rationalizes the individual SDGs selected in accordance to the formed SYDP SDGs Framework.

2.8.1 SDGs Identification and Justification

The proposed decentralized DC nano-grid system directly contributes to key Sustainable Development Goals (SDGs) through efficient energy generation, intelligent control, and community-level deployment.

Those directly addressed SDGs are as follows:



Figure 1.1 SDGs

According to the SYDP SDGs Framework, the selected goals for this project are:

SDG-9: Industry, Innovation, and Infrastructure (Mandatory Baseline — NCRC)

This project addresses SDG-9 through the design of sophisticated control systems (P&O MPPT; ILP-based CEMS) integrated across PCB hardware, embedded firmware, and Firebase cloud — forming a scalable, decentralized cyber-physical system that harmonizes academic theory with industrial reality as mandated by NCRC.

SDG-7: Affordable and Clean Energy (Primary Goal)

This project addresses SDG-7 by delivering a low-cost solar electrification solution for rural areas experiencing 10–14 hours of daily load shedding, using MPPT-based harvesting, BESS management, IoT monitoring, and ILP load scheduling — all built from affordable off-the-shelf components accessible to low-income households.

SDG-8: Decent Work and Economic Growth (Secondary Goal)

By supplying dependable 12V DC power to homes and small businesses suffering up to 14 hours of daily load shedding, this project extends productive economic activity and ensures continued access to lighting, digital services, and small business operations through intelligent load prioritization.

SDG-11: Sustainable Cities and CommunitieS (Secondary Goal)

This project promotes SDG-11 through IoT-based smart energy infrastructure — ESP32, Firebase, and web dashboard — giving residents real-time visibility and remote control over energy use, while swarm peer-to-peer sharing and dual-state load control build community resilience and autonomous energy governance.

SDG-12: Responsible Consumption and Production (Secondary Goal)

The ILP-based CEMS minimizes unnecessary battery drainage across all 32 load combinations, the MPPT algorithm ensures no solar energy is wasted, and hard SoC limits of 14.4V and 11.0V protect battery health — collectively embodying SDG-12's responsible consumption mandate.

SDG-13: Climate Action (Secondary Goal)

The Swarm DCNG eliminates diesel generator dependence in rural off-grid communities by providing a fully solar-powered system with battery backup, where the control system maximizes solar deployment before drawing from batteries — delivering pilot-scale climate action aligned with SDG-13.

3 System Design

3.1 System Architecture and Design

A decentralized Swarm DC Nano Grid is proposed to facilitate effective generation, storage, and sharing of power among multiple households. Each node operates independently while being interconnected by a common DC bus and is thus able to perform bidirectional energy exchange. The generation part of the proposed system incorporates photovoltaic generation and battery storage as well as power electronic converters to ensure efficient energy generation and management. The monitoring aspect is handled through the implementation of an IoT-based monitoring system.

The overall system architecture, including the interconnection of two DCNGs, is illustrated in Figure 3.1.

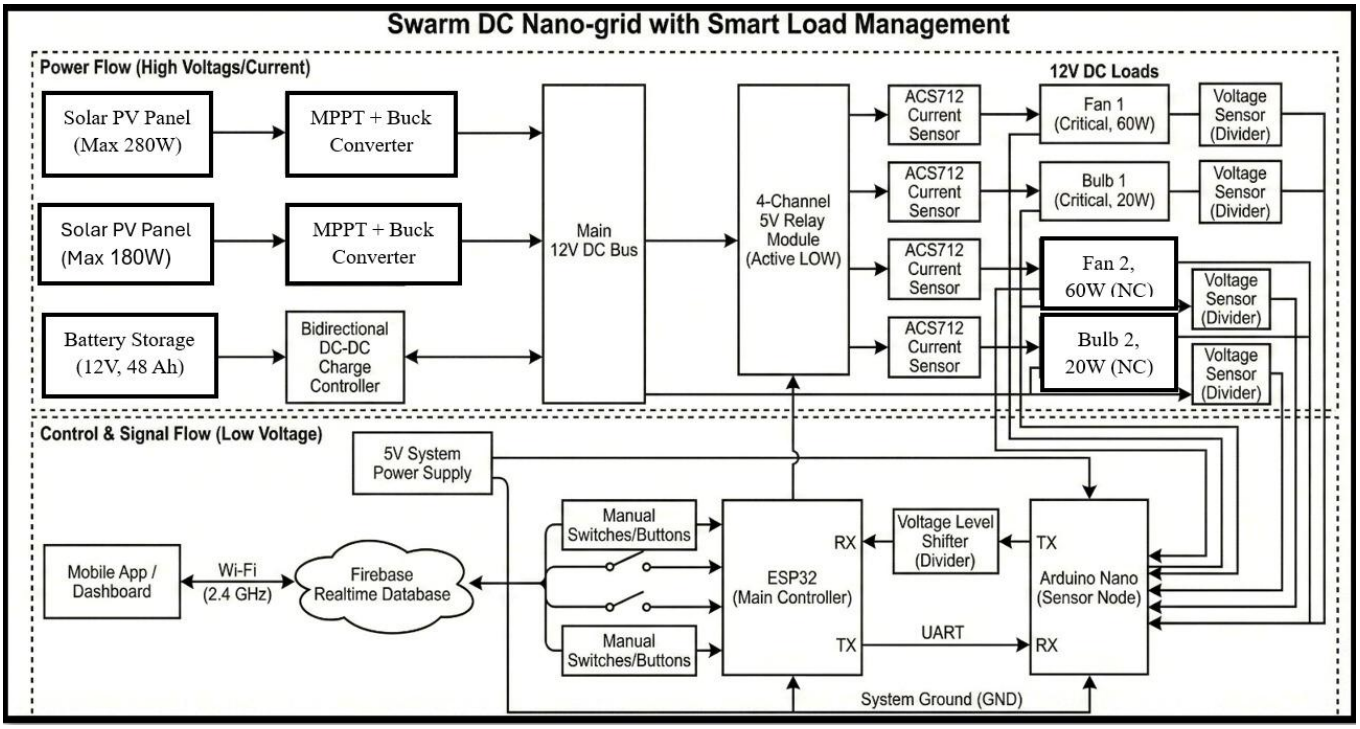


Figure 3.1 Proposed Architecture

Every DCNG node comprises a PV array, synchronous buck converter which also embeds the MPPT control, BESS which is interfaced through a bidirectional buck-boost converter, dual-microcontroller sensing and control system. Nodes are connected through common DC bus where the energy is exchanged bidirectionally, controlled through the cloud-based Cloud Energy Management System (CEMS).

The control and communication hierarchy is implemented at three levels:

- Device level: Arduino Nano is responsible for real time analog sensing and for the generation of MPPT duty cycles.
- Edge level: ESP32 collects the sensor data, manages the relay-controlled switching of loads and synchronization of cloud through the wireless networks.
- Cloud level: Python-based CEMS daemon constantly interrogates the Firebase Realtime Database to perform the ILP optimization and write the control commands back to edge devices.

3.2 DC Nano Grid Design

Each DC Nano Grid (DCNG) is composed of several key components that collectively enable energy generation, conversion, storage, and distribution. Following is its design in MATLAB-Simulink:

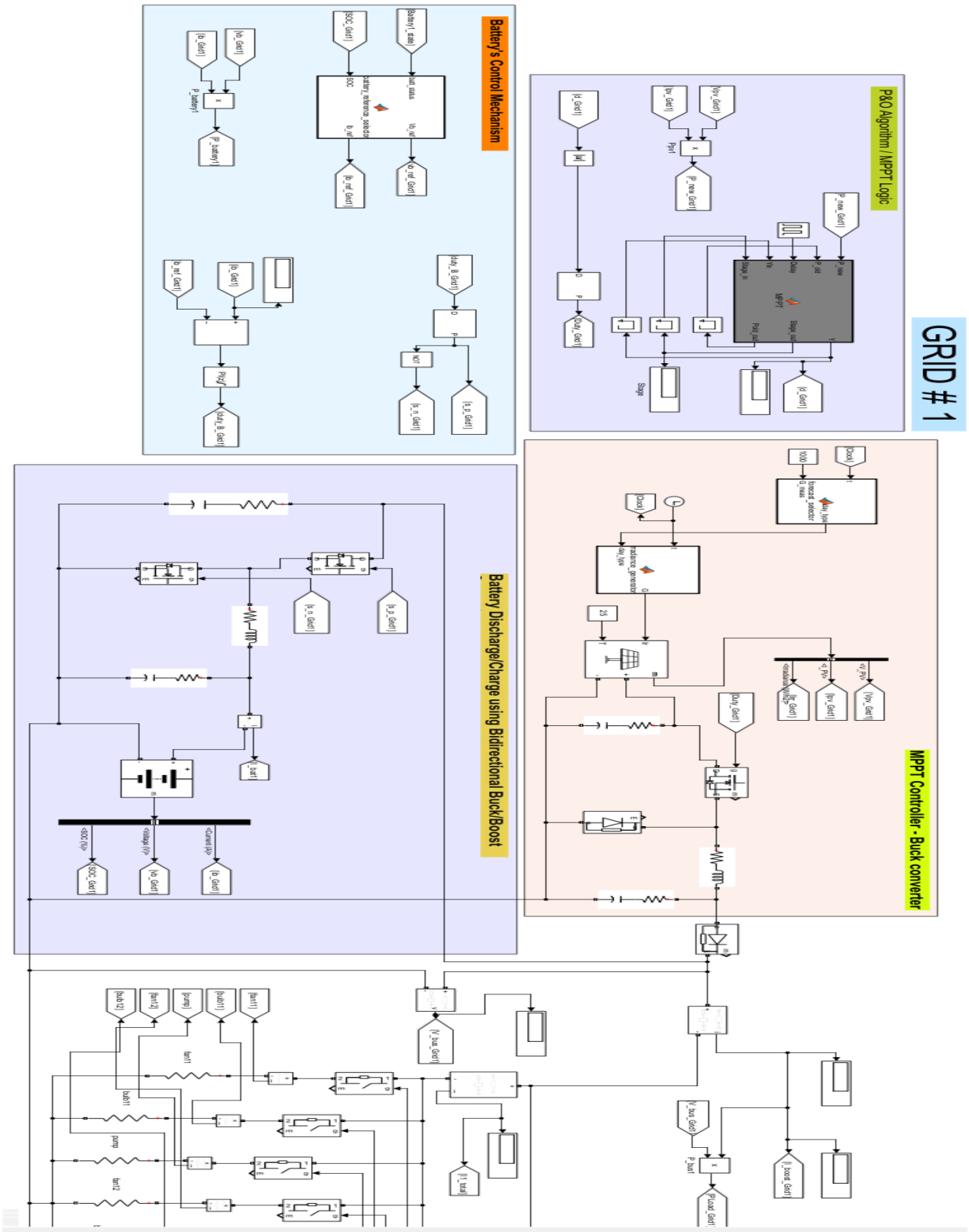


Figure 3.2 DC Nano Grid (DCNG)

3.2.1 Photovoltaic (PV) Array

The main source of energy in the nano grids is from photovoltaics. In the current study, the photovoltaic panel used in Grid 1 is the Mono's 280M-72, which has a power capacity of 280 W. The photovoltaic panel used in Grid 2 is the BK Solar XS-180-39M, which has a capacity of 180 W.

The following table lists the parameters of the PV panels used in both the grids:

Table 1.2 PV Panel Specifications

Parameters	**Specifications/Grid 1**	**Specifications/Grid 2**
Module Type	280M-72	XS-180-39M
Maximum Power - Pmax (W)	280	180
Open Circuit Voltage - Voc (V)	26.70	26.50
Short Circuit Current - Isc (A)	13.97	9.12
Maximum Power Voltage – Vmp (V)	22.22	23.01
Maximum Power Current – Imp (A)	11.82	8.71

3.2.2 MPPT with a Buck Converter

The buck converter is connected to the PV panel such that the regulated output is stepped down to the DC bus voltage of 12V. The buck converter is designed to have MPPT-P&O control scheme for maximization of the energy transfer from the PV cell to the load.

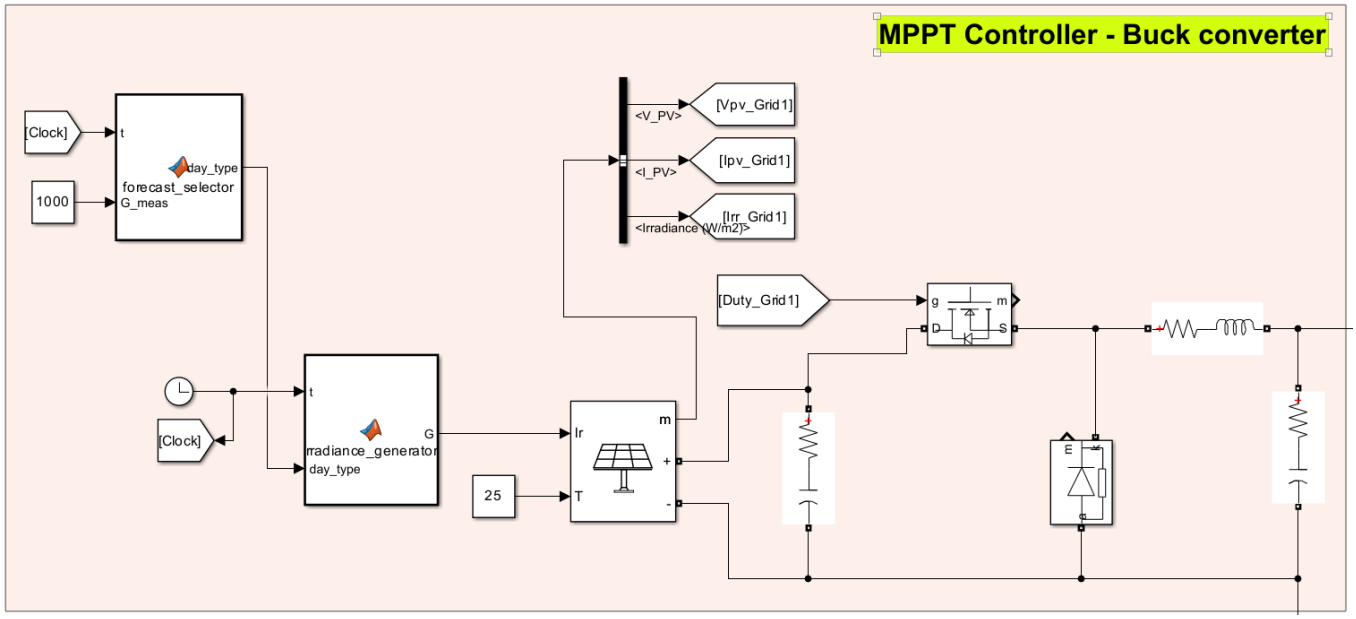


Fig. 3.3 MPPT interfaced with a buck converter

3.2.3 Irradiance Generator & Seasonal Profiles

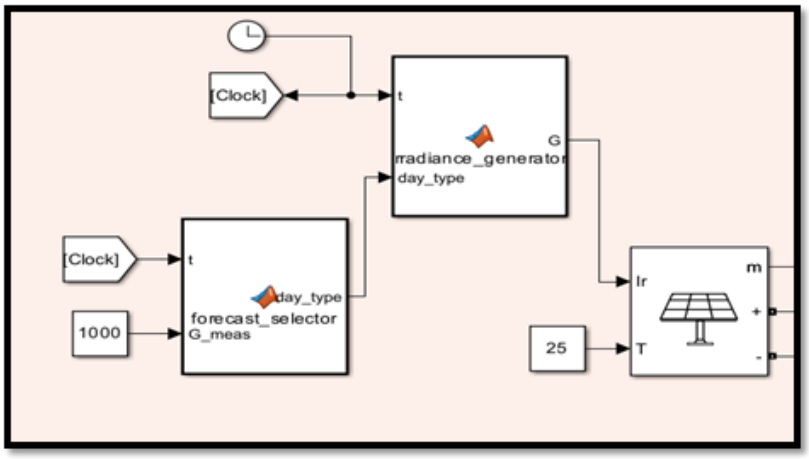


Fig. 3.4 Irradiance Generator

In order to assure that the simulations are executed in real working conditions, a function was created to create real values of irradiance from Topi, considering the measurements that are performed in the different seasons, as input for the model of the PV panel. It was taken the maximum value of the irradiance profile for each season because it is going to characterize the season. These values of irradiance profile for each season (summer, winter, cloudy and average season) are the following ones:

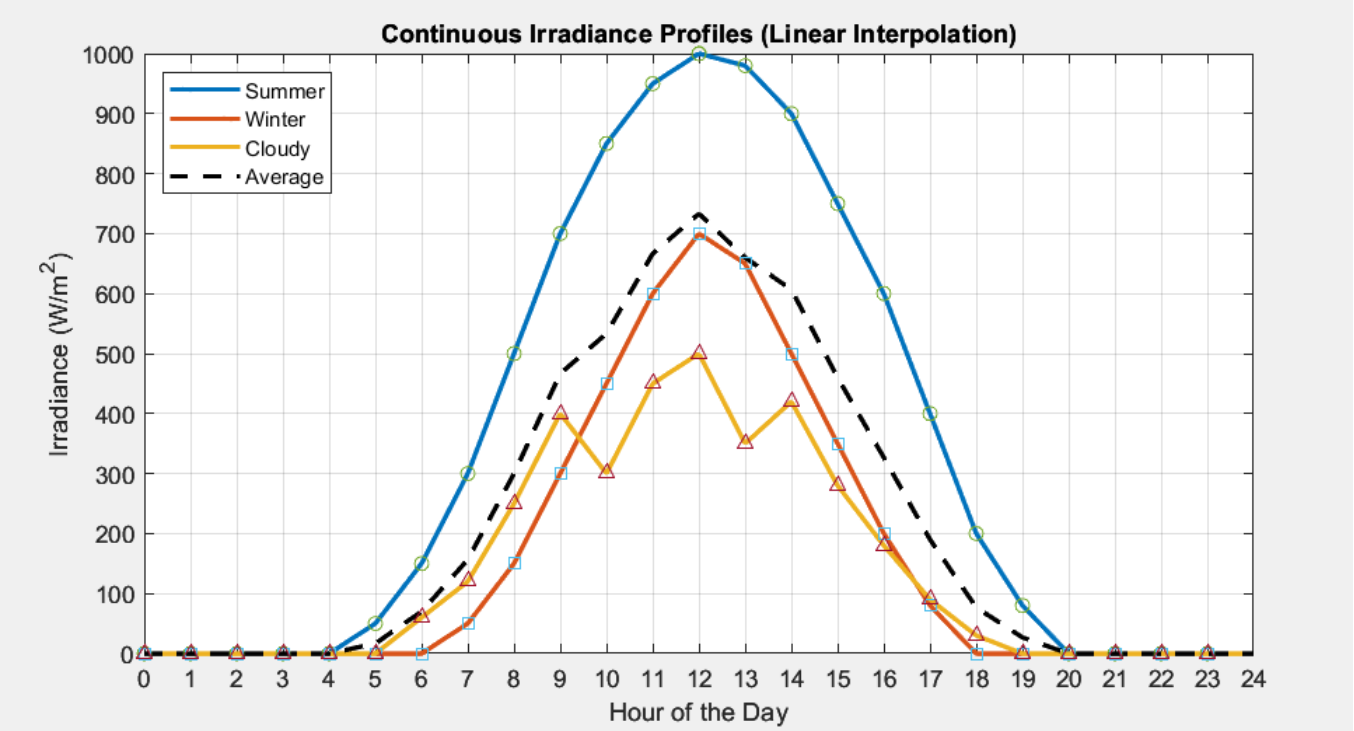


Fig. 3.5 Seasonal Solar Irradiance Profiles

3.2.4 Battery Energy Storage System (BESS)

In this project, the battery bank having 48 Ah 12 V capacity has related to the bidirectional buck boost converter which acts as the Battery Energy Storage System (BESS). The bidirectional buck boost converter will work as buck when the battery is charging from generation and works as boost when the battery is discharging to the DC Bus.

The following table lists the parameters of the shared battery used for both the grids:

Table 1.3 Battery Specifications

Parameters	**Specifications**
Nominal Voltage (V)	12

Nominal Capacity (Ah)	48
Type	Sealed Lead-Acid (Maintenance free)
Maximum Charging Current (A)	12.5 - 15
Internal Resistance (mΩ)	7 – 8 (when fully charged)
Cyclic Charging Voltage (V)	14.4 to 14.9 @ 25°C
Float Charging Voltage (V)	13.5 to 13.8 @ 25°C
Operating Temperature (°C)	Charge: 0 to 50
	Discharge: -20 to 60
Expected Life (years)	10 - 12

3.3 Battery Control system:

Battery charging and discharging are handled by a bidirectional converter regulated by a PI controller. The system keeps the DC bus voltage stable while ensuring safe battery operation within defined limits.

The whole system works in three states depending on the controller as follow:

Mode 1: Generation > load on DC Bus

Extra power is used to charge the battery or exchanged with other DCNG's by means of DC Bus battery status = +1 is set and battery reference selector selects battery reference=14.4 V.

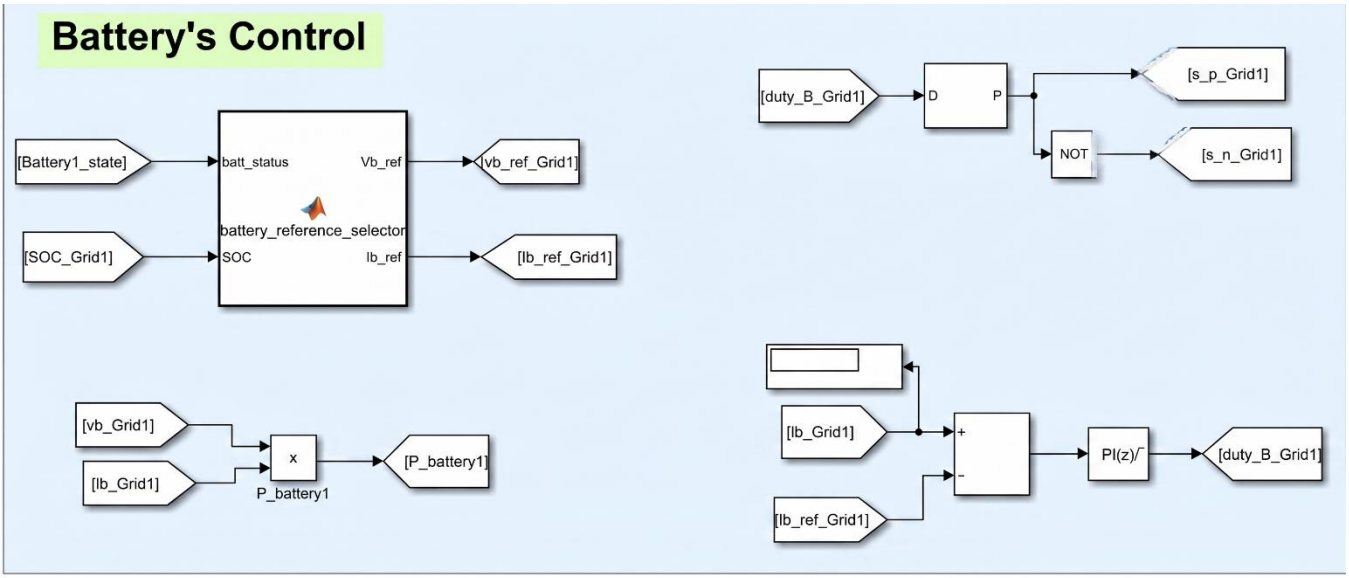
Mode 2: Generation = load on DC Bus

No exchange of power between the system and other parts. Battery status=0 is set and current reference=0 A is used to maintain battery idle.

Mode 3: Generation < load on DC Bus

Extra power needed by the load is from either battery or other DCNG's battery status = -1 is set and battery reference selector selects battery reference=11.0 V.

The charging voltage of battery is capped to 14.4 V, charging current to 5A (0.1 C) and the discharging voltage is to 11.0 V for protection against deep discharge.




```
batt_status = +1  → Vb_ref = 14.4V  (charging reference)
batt_status = -1  → Vb_ref = 11.0V  (discharging reference)
batt_status =  0  → Ib_ref = 0A     (idle, no current)
```

Figure 3.6 Battery Control Mechanism

3.4 Swarm DC Bus and Multi-Node Integration

All the DCNG nodes are coupled with a single high-current 12 V DC bus through which the energy can flow from or to other DCNG nodes. Each DCNG node interfaced with the 12V bus can inject the excess energy into the bus or draw energy from the bus via a bi-directional DC-DC converter based on the status of SoC and the load condition. The coordination layer of swarm is taken care by the cloud-based CEMS which oversees all the DCNG nodes in real time, mediates sharing of energy and enables making optimal decisions globally whereas the existing multi-node nano-grid architectures follow a fixed or rule-based energy sharing methodology.

3.5 Converter Design and Implementation

3.5.1 Synchronous Buck Converter

A synchronous buck converter is designed to step down the input voltage from the PV system (26.7 V from the 280 W panel while 23.01 V from a 180 W panel) to the required DC bus voltage of 12 V. The key design parameters include inductor selection, capacitor sizing, switching device characteristics, and duty cycle optimization.

This topology is employed to improve efficiency by replacing the conventional diode with a controlled MOSFET, thereby reducing conduction losses. This is particularly beneficial in low-voltage DC systems where efficiency improvements significantly impact overall system performance.

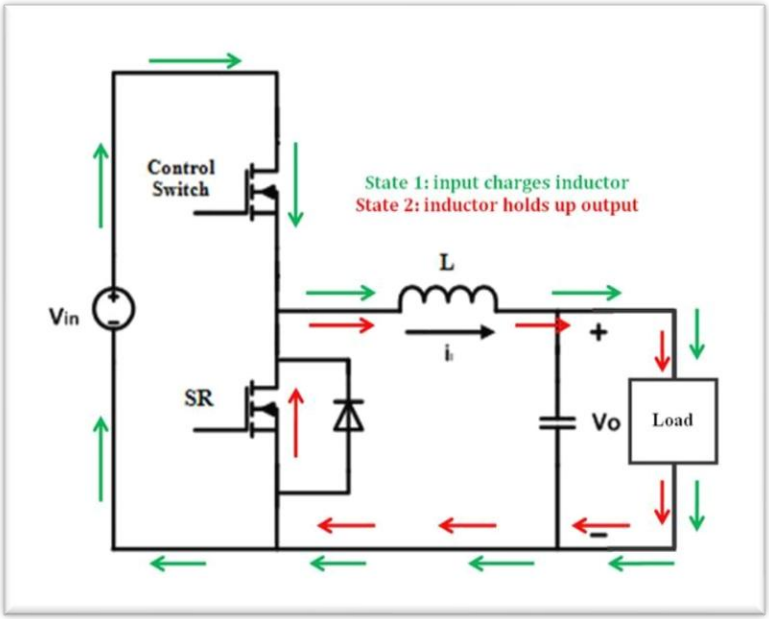


Figure 3.7 Synchronous Buck Converter

For an input of 24 V and output of 12 V, the duty cycle is calculated as 0.5, indicating that the high-side switch operates for 50% of the switching period. The duty cycle D is determined using:

$$\boldsymbol{D = \frac{V_{out}}{V_{in}} = \frac{12}{24} = 0.5}$$

Assuming $\boldsymbol{f_s = 100\ \text{kHz},\ \Delta I = 0.6\ \text{A}}$:

The inductor value is selected to ensure continuous conduction mode (CCM) and controlled current ripple with an approximate inductance of:

$$\boldsymbol{L = \frac{(V_{in} - V_{out}) \cdot D}{\Delta I_L \cdot f_s} \approx 100\ \mu H}$$

The output capacitor is selected to limit voltage ripple with an approximate capacitance of:

$$C = \frac{\Delta I_L}{8 \cdot f_s \cdot \Delta V_{out}} = 62.5\ \mu F \approx 68\ \mu F$$

A typical 68F low-ESR capacitor is chosen. To minimize the conduction losses and improve efficiency, a synchronous design with two N-channel MOSFETs is chosen. A good design of gate driver with accurate dead time control is essential for safe and efficient switching. The designed converter can produce a stable 12V output with low output ripple and high efficiency.

3.5.2 Bidirectional Buck-Boost Converter

The bidirectional buck–boost converter is designed to enable controlled energy exchange between the battery energy storage system (BESS) and the DC bus. It operates in both buck and boost modes depending on system conditions, ensuring stable voltage regulation and efficient power flow.

Topology: MOSFET-based half-bridge configuration

Modes of Operation:

- **Buck Mode:** Battery discharging
- **Boost Mode:** Battery charging

Control: PI-based voltage and current regulation

This converter ensures stable bus voltage and seamless transition between operating modes.

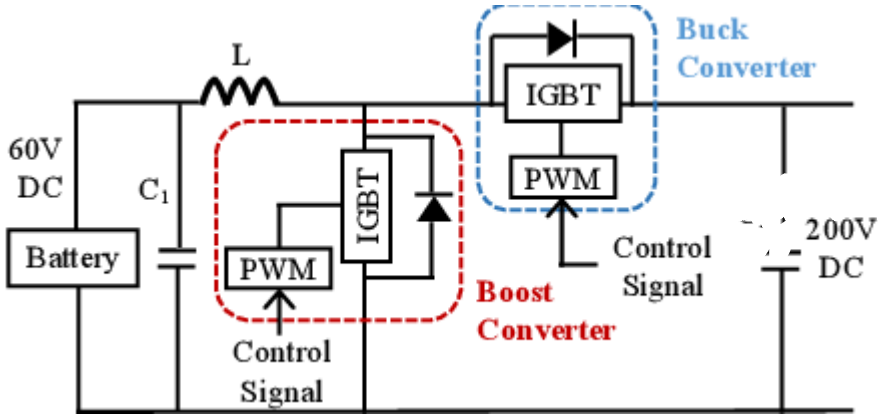


Figure 3.8 Bidirectional buck-boost Converter

The duty cycle varies based on the mode of operation:

1) **Buck Mode (Battery Discharging):**

When the battery supplies power to the DC bus, the converter operates in buck mode. A fully charged 12 V, 48 Ah lead-acid battery gives around 12.6 – 12.8 V.

The duty cycle is given by:

$$\boldsymbol{D = \frac{V_{out}}{V_{in}} = \frac{12}{12.7} \approx 0.945}$$

2) Boost Mode (Battery Charging):

When excess solar power is available, the battery charges. Since the required charging voltage (≈14.4 V) is higher than the DC bus voltage (12 V), the converter operates in boost-assisted mode rather than pure buck operation.

When the battery is supplied with power from the DC bus through excess solar generation, the converter operates in boost mode. A 12 V, 48 Ah lead-acid battery needs around 13.5 – 13.8 V to be charged. The duty cycle is given by:

$$\boldsymbol{D = 1 - \frac{V_{in}}{V_{out}} = 1 - \frac{12}{13.65} \approx 0.121}$$

Assuming $\boldsymbol{f_s = 100\ \text{kHz},\ \Delta I = 0.6\ \text{A}}$**:**

The inductor value is selected to maintain continuous conduction mode (CCM) and limit current ripple with an approximate inductance of:

$$\boldsymbol{L = \frac{V_{in} \cdot D}{f_s \cdot \Delta I} \approx 100\ \mu H}$$

The output capacitor is selected to limit voltage ripple with an approximate capacitance of:

$$\boldsymbol{C = \frac{I_{out} \cdot D}{f_s \cdot \Delta V} \approx 68}\ \mu\text{F}$$

3.6 MPPT Control Algorithm

The Perturb and Observe (P&O) algorithm is implemented for real-time tracking of the maximum power point. It adjusts the duty cycle based on incremental changes in power, ensuring optimal energy harvesting.

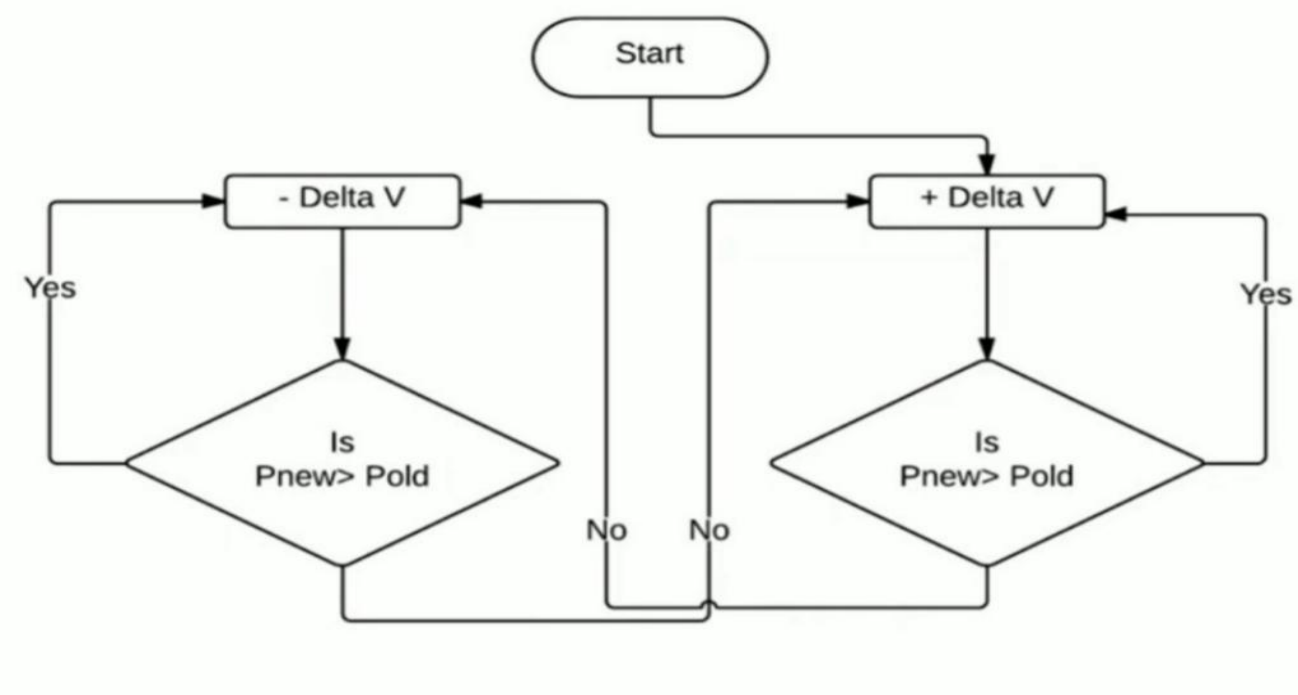


Figure 3.9 P&O flowchart for MPPT

The flowchart represents the Perturb and Observe (P&O) algorithm, a widely used Maximum Power Point Tracking (MPPT) technique for photovoltaic (PV) systems. The method operates by applying a small perturbation to the PV voltage and observing the resulting change in output power.

The fundamental relationship is:

$$\boldsymbol{P = V \cdot I}$$

where (P) is power, (V) is voltage, and (I) is current.

At each iteration, the algorithm evaluates:

$$\boldsymbol{\Delta P = P_{new} - P_{old}}, \qquad \boldsymbol{\Delta V = V_{new} - V_{old}}$$

This behaviour is based on the slope of the power-voltage curve:

$$\boldsymbol{\frac{dP}{dV} \approx \frac{\Delta P}{\Delta V}}$$

The tracking decision is based on the sign of $\frac{dP}{dV}$:

- $\frac{dP}{dV} > 0 \rightarrow$ Operating point is to the left of the MPP; increase voltage (decrease duty cycle).
- $\frac{dP}{dV} < 0 \rightarrow$ Operating point is to the right of the MPP; decrease voltage (increase duty cycle).
- $\frac{dP}{dV} = 0 \rightarrow$ Operating point is at the MPP; maintain current duty cycle.

3.7 Cloud Energy Management System (CEMS) and ILP Optimisation

The Cloud Energy Management System (CEMS) is a daemon written in Python which is executed on a cloud server. The system continuously watches the Firebase Realtime Database and gets the current system parameters, performs the ILP optimisation, and writes the desired load commands into Firebase where the ESP32 at the edge device executes them.

3.7.1 Optimisation Objective and Cost Function

The ILP optimiser evaluates all possible binary on/off states for the five connected relay loads every 2 seconds, selecting the configuration that minimises the following cost function:

$$J = \sum_i \left(W_i \cdot S_{\text{shed},i}\right) + \alpha \cdot | P_{\text{batt}} |$$

Variable Definitions

- J - Total cost function to be minimized by the ILP optimizer. It represents the overall penalty of a given load configuration.
- i - Index representing each load (e.g., Fan 1, Lights, Fan 2 etc.).
- W_i - (Weight / Shedding Penalty): Penalty assigned to shedding (turning OFF) load i.
  High value for critical loads (e.g., 1000)
  Low value for non-critical loads (e.g., 10)
- $S_{\text{shed},i}$ - *(Shedding State)*: Binary variable indicating whether load i is shed.
  1: Load is OFF (shed)
  0: Load is ON
- α - *(Battery Usage Penalty Coefficient)*: A tuning parameter that penalizes the use of battery power, encouraging preference for solar energy.
- P_{batt} - *(Battery Power):* Total power drawn from the battery (in watts).

Decision Logic

- Shedding Penalty (W): Shedding a user-requested critical load (e.g., Fan 1, Lights) incurs a very high penalty (1000 points), while shedding non-critical loads (e.g., Fan 2) incurs a smaller penalty (10 points).
- Battery Wear Penalty (α): Using battery power instead of solar power introduces an additional cost, encouraging the system to prioritize direct solar usage whenever sufficient irradiance is available.

3.7.2 The Constraints Engine

- **Constraint A (User Intent):** The algorithm never activates a load unless it has been explicitly requested by the user through the dashboard or physical toggle switches $\mathbf{req_relay}$.
- **Constraint B (Battery Viability):** The total required battery discharge power $\mathbf{P_batt_needed}$ must not exceed the system's maximum safe discharge limit (60 W / 5 A for the 12 V system).
- **Constraint C (Critical Protection):** If the State of Charge (SOC) drops to ≤ 40%, the maximum discharge power is forced to 0 W, enforcing a hard cutoff to protect the lead-acid battery from deep-discharge sulfation. If SOC is < 60%, battery discharge is penalized to encourage critical charging.

The following figure shows the ILP-based Cost function implementation:

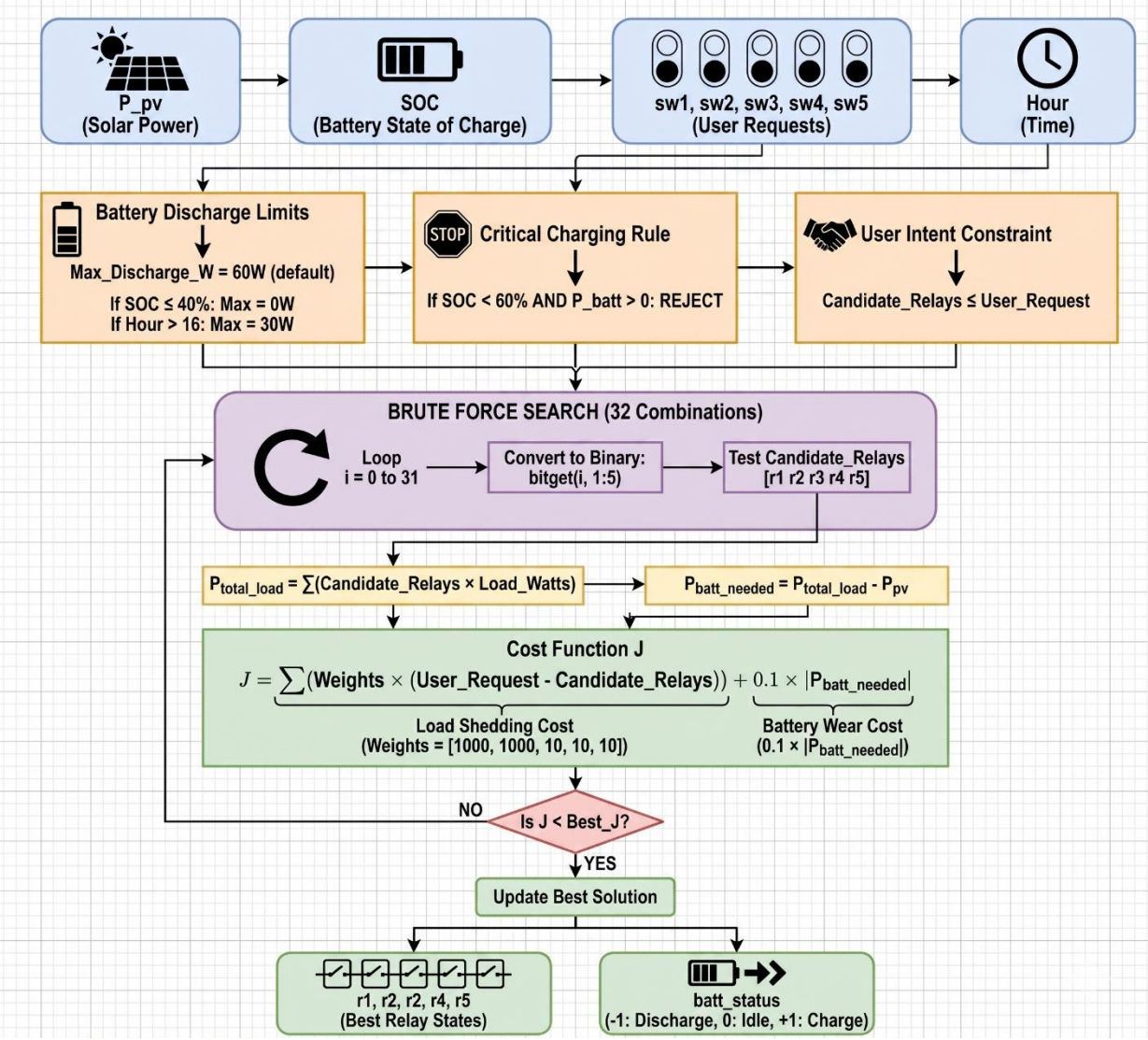


Figure 3.10 ILP-Based Cost Function and Load Optimisation Decision Flow

3.8 Web-Based Dashboard and User Interface

The developed web-based dashboard provides an interactive interface for real-time monitoring and control of the DC Nano Grid system. It is designed using HTML, CSS, and JavaScript, with graphical visualization implemented through Chart.js. The dashboard communicates with the cloud database to display live system parameters and execute control commands.

3.8.1 Real-Time System Monitoring

The dashboard displays key electrical parameters including:

- Solar Generation (Voltage, Current, Power): Indicates real-time PV output and system generation capability under varying irradiance conditions.
- Load Consumption: Shows the total power demand of connected loads, enabling users to observe consumption patterns.
- Battery Status (Voltage, Current, SoC): Provides insight into battery health and operating mode (charging/discharging).

Graphical plots are used to visualize these parameters over time, allowing users to analyse trends, detect fluctuations, and evaluate system performance under dynamic conditions.

3.8.2 User Control and Dual-State Logic

The dashboard includes toggle controls for connected loads (e.g., fans, lights, pump), allowing users to request activation or deactivation remotely. However, the system operates on a dual-state logic:

- User Request State: Represents the user's desired action
- Actual System State: Reflects the final decision after cloud-based optimization

This ensures that user commands are subject to system constraints such as available solar power and battery SoC, preventing unsafe or inefficient operation.

4 Hardware Implementation

This chapter explains the implementation of the DCNG system in physical hardware: PCB design, component selection, IoT infrastructure, and dual MCU sensor/controller architecture. The simulation-proven designs developed in Chapter 3 are implemented.

4.1 PCB Designing & Fabrication

For every primary module within the system a custom PCB was designed and manufactured. The boards were designed to ensure compact integration, electrical robustness and efficient power delivery. Every board includes over current protection using detection and shutdown logic, over voltage protection using clamping and control shut down schemes, and transient suppression filters to prevent damage to critical components.

4.1.1 MPPT Power Stage with Buck Converter

Two individual MPPT PCBs were designed, one for each DCNG node. Each board utilizes the synchronous buck topology based on high-frequency MOSFET switching, a low-power optimized gate driver with controllable dead time and inductive components with appropriate ratings. Layouts design is taken care with the provision of low inductance paths for high currents, star grounding to eliminate potential difference between grounds and thermal relief pads. The manufactured boards are presented for Grid 1 in Figure 4.1 and for Grid 2 in Figure 4.2 respectively.



Figure 4.1 Grid 1: MPPT and a buck converter

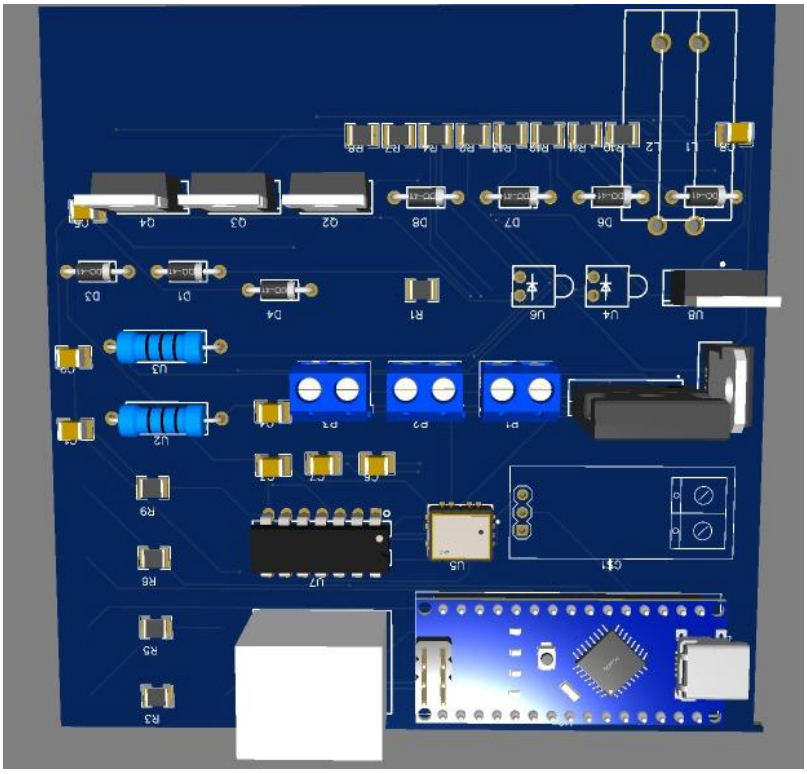


Figure 4.2 Grid 2: MPPT and a buck converter

4.1.2 Bidirectional Buck-Boost Converter for Battery Management

This PCB realizes a non-isolated bidirectional DC-DC topology enabling controlled energy transfer between the battery and DC bus. A symmetric MOSFET arrangement supports both charging (buck mode) and discharging (boost mode) operations, governed by closed-loop control strategies. The design ensures seamless mode transition, high conversion efficiency, and stable operation under varying load and state-of-charge conditions.

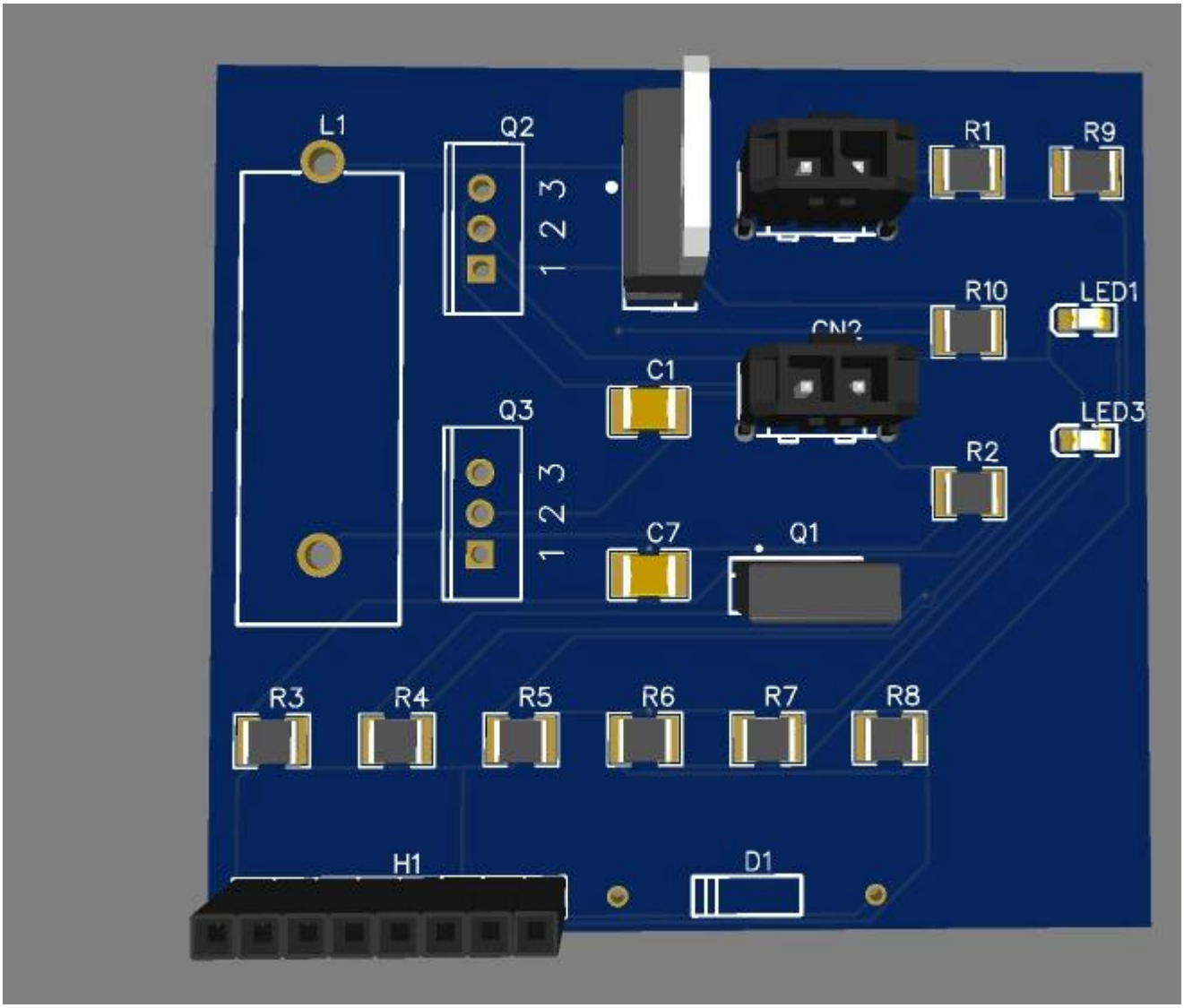


Figure 4.3 bidirectional buck-boost converter for battery controller

4.1.3 Sensor Interfacing and Signal Conditioning Circuitry

Dedicated PCBs are developed for accurate real-time acquisition of electrical parameters. These include precision voltage divider networks, filtering stages, and isolation considerations to ensure noise resilient measurement signals suitable for ADC interfacing.

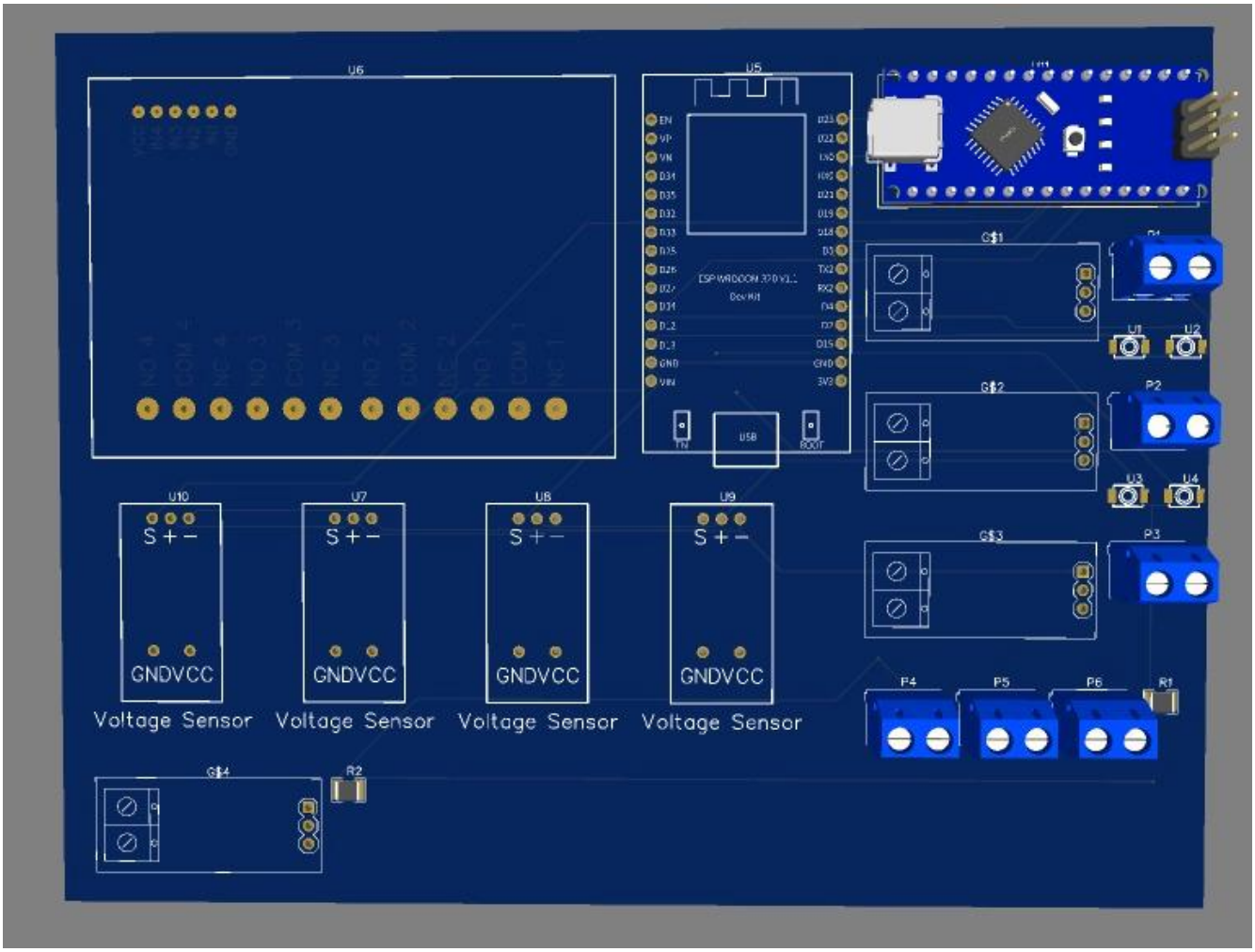


Figure 4.4 Sensing circuit

To enhance system reliability and operational safety, integrated protection mechanisms are implemented across all boards, including:

- Overcurrent protection via sensing and cutoff logic
- Overvoltage protection using clamping and control shutdown strategies
- Transient suppression and filtering to safeguard sensitive components

4.2 Dual-MCU Control and IoT Architecture

The hardware control system has a dual-microcontroller structure. It is intended to physically distance high fidelity analog sensing from computation-heavy wireless communication and cryptographic operations, to avoid CPU blocking and SSL memory collapse associated with using both on a single processor.

4.2.1 The Sensing Worker: Arduino Nano

The Arduino Nano is solely dedicated to the real-time data acquisition; its logic is at 5 V, its 10-bit ADC to read 4 input channels for measurements. Voltage is measured through a 5:1 resistive divider, while current is measured by ACS712 Hall effect sensors (V/A sensitivity: 0.066 V/A, V(0) offset: 2.5 V; 30 A module). To get rid of switching transient signals and high frequency noise of power converters, a moving average filter (SAMPLES=10) is implemented in firmware to smooth out measurements. The smoothed voltage and current readings are then compiled into a thin comma-separated telemetry string, delimited by a '#' character at start and '!' character at the end, and sent to the ESP32 at 9600 baud on a UART interface.

4.2.2 The IoT Manager: ESP32

The ESP32 WROOM-32 serves as the system's IoT gateway, operating at 3.3 V logic, where it parses UART telemetry received from the Arduino Nano, controls load switching through a 4-channel relay module, and synchronizes data with the Firebase Realtime Database over Wi-Fi. It interfaces with four AceButton-based physical toggle switches, but instead of directly actuating relays, each switch updates an "intent" flag in Firebase, allowing a cloud-based CEMS to authorize or deny the requested operation. Additionally, the ESP32 directly samples solar voltage and current using its ADC1 pins (GPIO 34 and GPIO 35), specifically chosen because ADC2 becomes unavailable during active Wi-Fi communication. To ensure safe voltage compatibility, the Arduino Nano's UART TX line is passed through a hardware voltage divider (2 kΩ / 1 kΩ) to step down the 5 V signal to a level suitable for the ESP32's RX pin.

The full hardware pinout matrix is given below:

Table 4.1 Hardware Pinout Matrix

Component / Function	Microcontroller	Pin / Interface	Notes
Relay 1 (Fan 1)	ESP32	GPIO 18	Active-LOW Control
Relay 2 (Lights)	ESP32	GPIO 19	Active-LOW Control
Relay 3 (Pump)	ESP32	GPIO 32	Active-LOW Control
Relay 4 (Fan 2)	ESP32	GPIO 5	Active-LOW Control
Switch 1 (Req 1)	ESP32	GPIO 13	Internal Pull-up
Switch 2 (Req 2)	ESP32	GPIO 12	Internal Pull-up
Switch 3 (Req 3)	ESP32	GPIO 14	Internal Pull-up
Switch 4 (Req 4)	ESP32	GPIO 27	Internal Pull-up
Solar Voltage (PV)	ESP32	GPIO 34 (ADC1)	via 5:1 Divider
Solar Current (PV)	ESP32	GPIO 35 (ADC1)	via ACS712
UART RX (from Nano)	ESP32	GPIO 16 (RX2)	via 3.3V Logic Divider

UART TX (to Nano)	ESP32	GPIO 17 (TX2)	Connected to Nano RX0

4.3 Firebase Realtime Database Architecture

Firebase Realtime database is used for bidirectional communication between ESP32 edge device and Python based CEMS. In the database structure the telemetry readings (Voltage, Current, power of four channels and solar values) and state of the relay commands are read and write separately by ESP32 and CEMS.

4.3.1 Optimised Data Upload Strategy

In our initial implementation, we transmitted 13 sensor readings in 13 individual Firebase.setFloat() commands. Each command created a separate SSL/TLS connection and as a result, we incurred connection overhead, wasted considerable RAM and soon ran out of Wi-Fi and the entire system crashed. This implementation was rewritten using the FirebaseJson object. This allowed us to bundle all 13 telemetry readings in a single JSON bundle and update a single node using one Firebase.updateNode() command. The number of SSL/TLS connection over heads was reduced by over 92% and this eliminated the system memory crashes.

4.3.2 Asynchronous Polling for Relay Commands

The initial continuous stream subscription for the relay commands led to unexpected loops where the uploaded telemetry would then come back down the same path in the database. This ate up CPU time, resulting in skipped commands. The implemented approach uses an asynchronous poll method in which the ESP32 asks Firebase for relay states every 1.5 seconds using Firebase.getBool(). This approach separates uploads from downloads and provides reliable deterministic control of the relays. Figure 4.5 displays the Firebase database in action during system operation.

The following figure shows the ESP32 controlled 4-channel relay module’s circuit connection diagram:

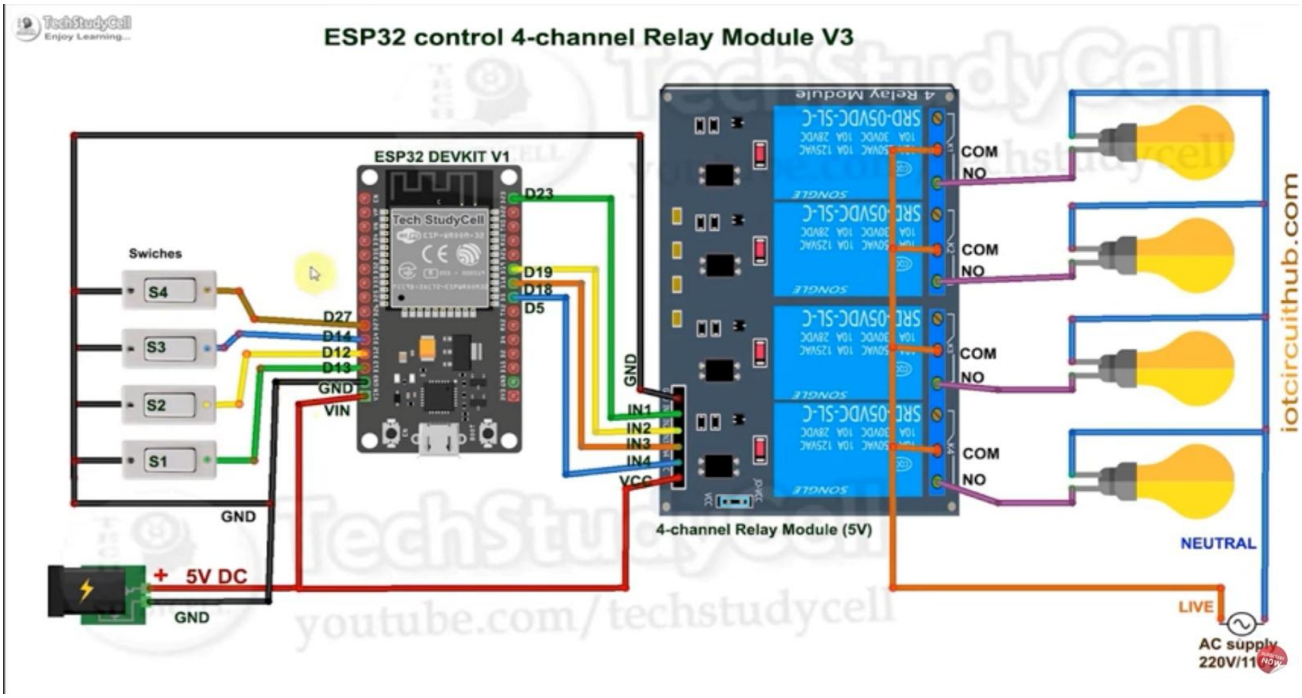


Figure 4.5 ESP32 controlled 4-channel relay module

The following image shows the data being received from the sensors on Google Firebase:

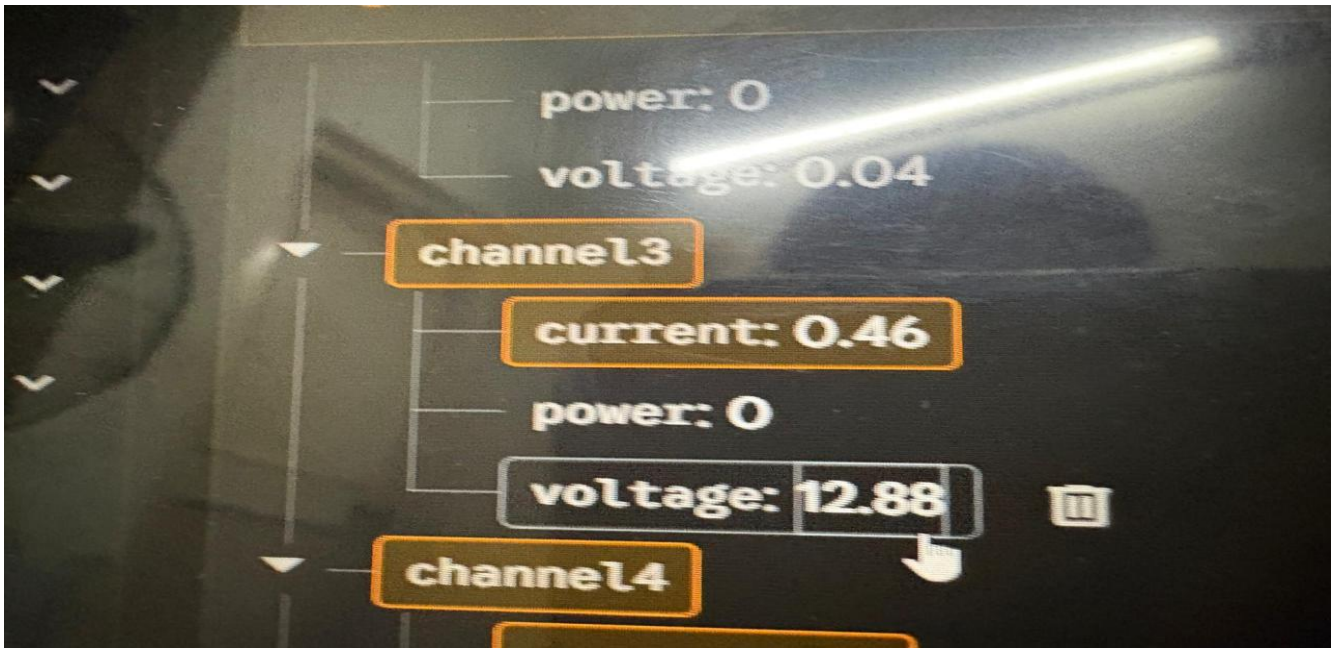


Figure 4.6 Google firebase database

4.4 System Component Integration

The interaction between all hardware components (sub-systems) is facilitated through common 12 V DC bus and common signal ground. Assembly order during prototype integration:

1. PV panel (280 W nominal for Grid 1; 180 W nominal for Grid 2) linked to MPPT buck converter PCB.

2. DC bus (12 V) established, carrying copper bus bars to support current.

3. Battery (12 V; 48Ah) connected through bidirectional buck-boost converter PCB.

4. Sensing circuit PCB placed across the load channels and across the battery terminals.

5. Arduino Nano interfaced to the sensing PCB and used for collecting ADC values.

6. ESP32 integrated with UART connection to Arduino Nano, the relay module and Wi-Fi for Firebase connection.

7. 4-channel relay module in place to control the load switching.

The following images are the assembled hardware components:

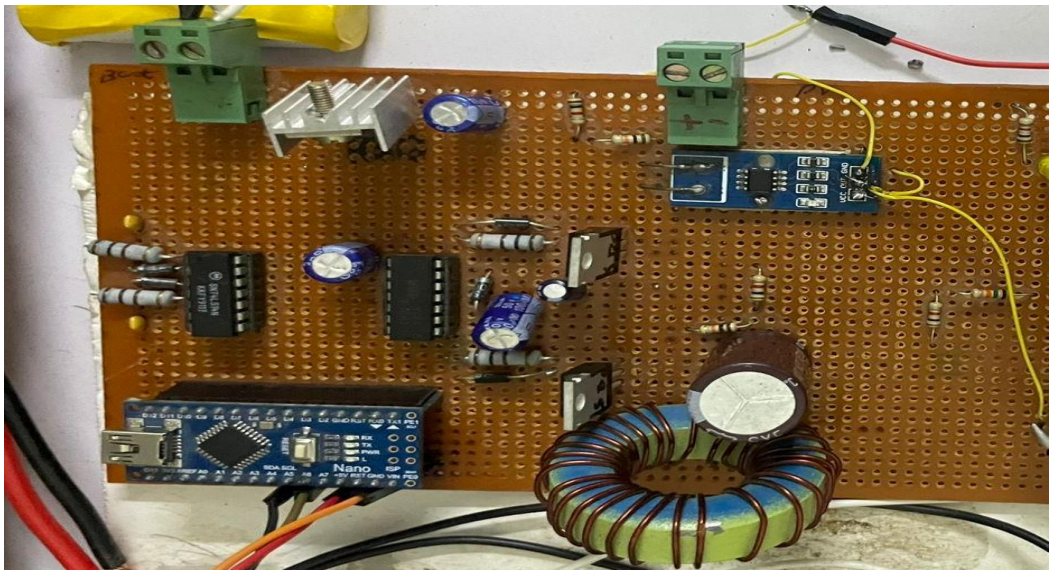


Figure 4.7 Fabricated MPPT Circuit

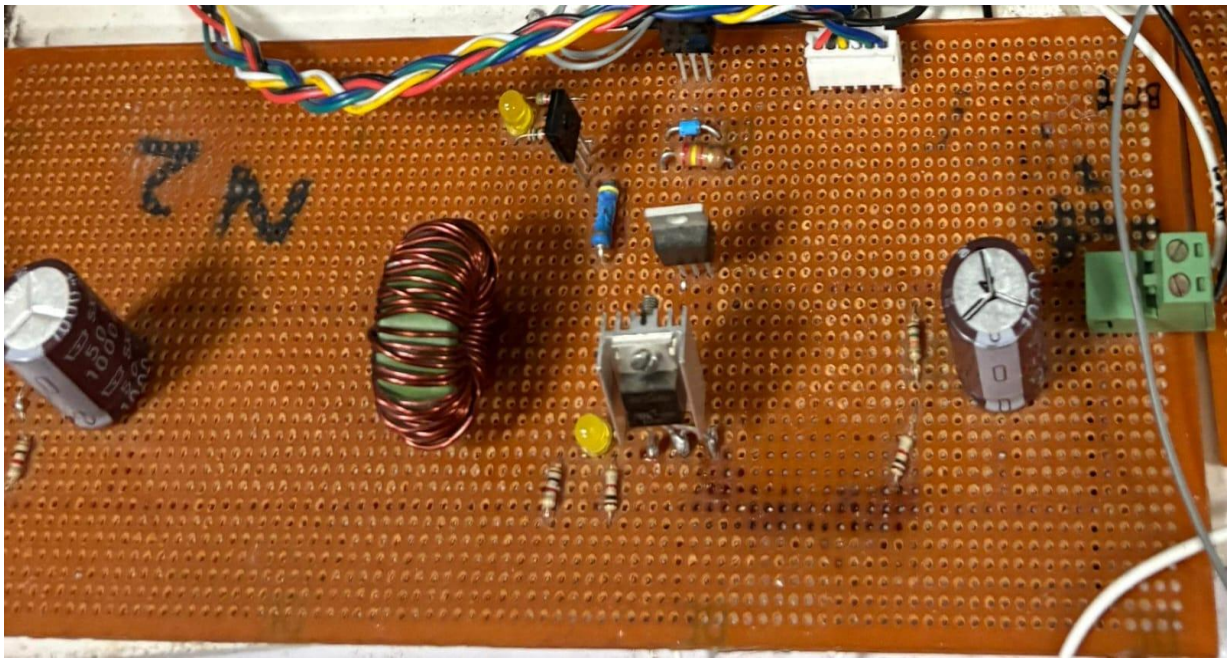


Figure 4.8 Fabricated Bidirectional Converter

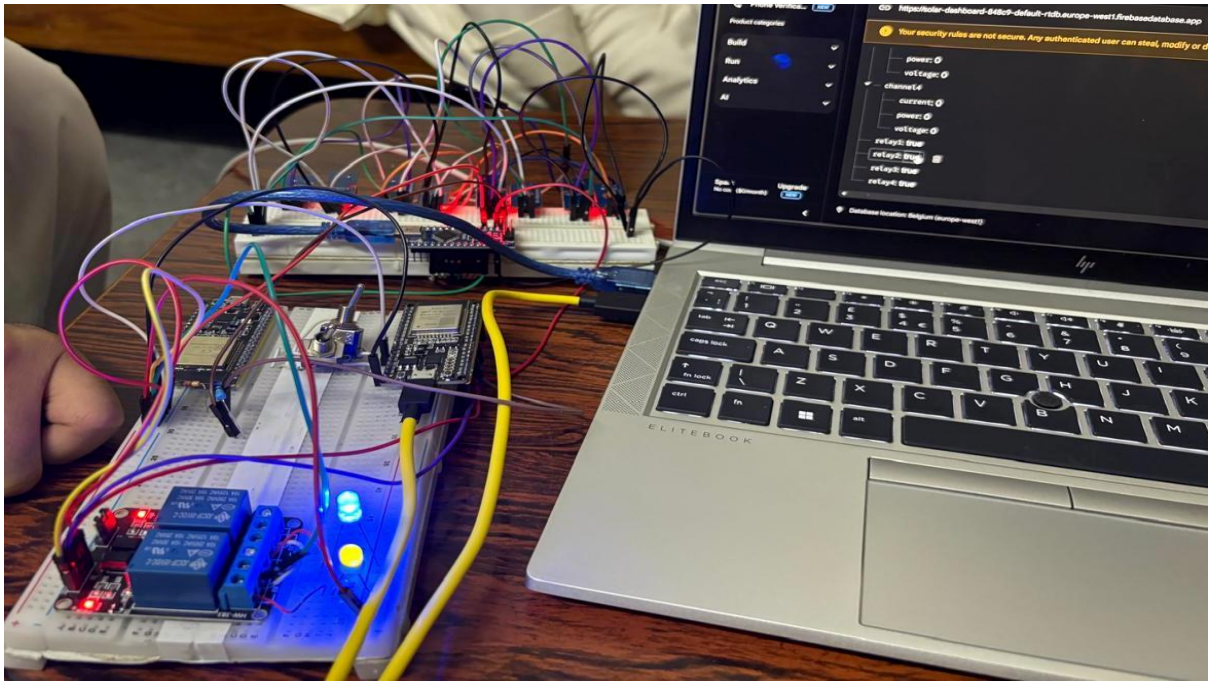


Figure 4.9 Sensor and ESP32 Control Circuit

5 Simulation Results and Analysis

This chapter shows the results from the MATLAB/Simulink simulation campaign used for validating the design of the single DCNG node before physical implementation. All simulations are run using the summer solar irradiance trace obtained at Topi, KPK, with temperature held constant at 25C, except where specified.

5.1 Simulation Configuration

A detailed simulation of a single DC Nano Grid (DCNG) node was carried out using MATLAB/Simulink to evaluate baseline system performance. The following parameters were used:

- PV Panel Capacity: ~280 W, 180 W
- Load Demand: ~160 W [critical= 80, non-critical=80]
- Battery Capacity: 12 V, 48 Ah
- DC Bus Voltage: 12 V

5.2 PV System Performance Under Variable Irradiance

PV subsystem was tested with the summer irradiance profile to reflect a real environment. Figures 5.1 show the time response of solar irradiance, PV current, PV voltage and retrieved PV power over one day. Figures 5.1 show the algorithm successfully tracked the MPP with a fluctuating irradiance curve and followed the solar irradiance curve of PV power output.

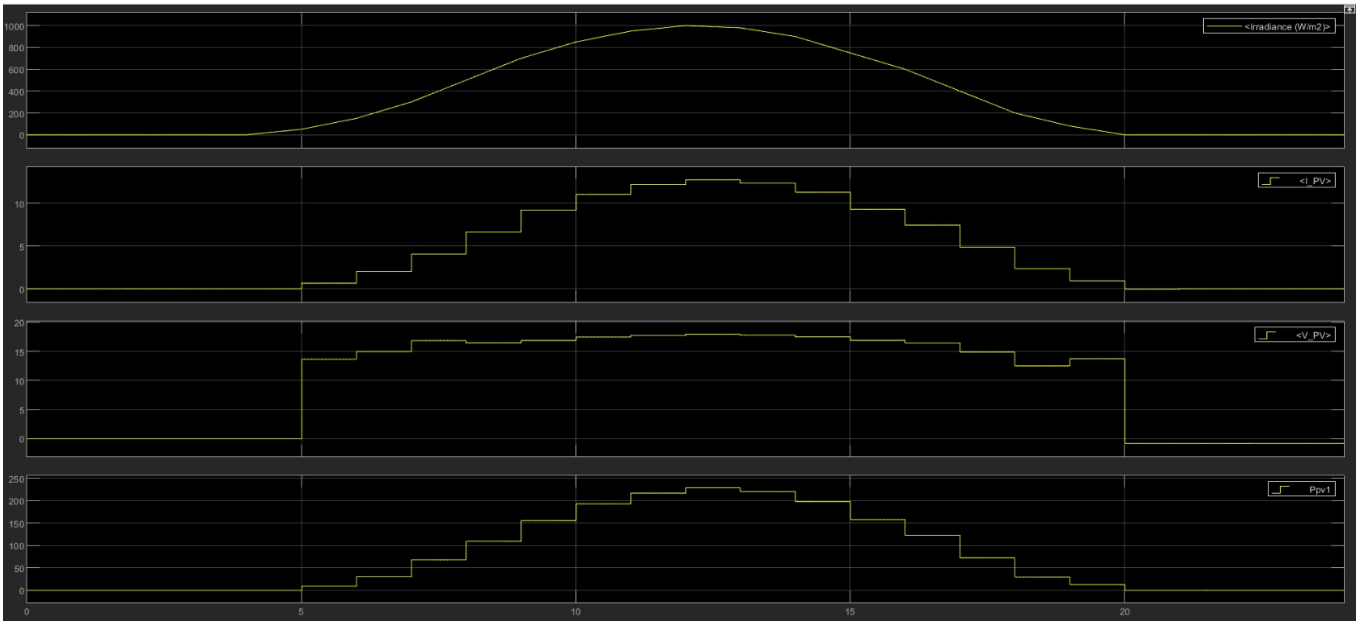


Figure 5.1 Summer Irradiance, Voltage, Current and Power of the PV panel in a DCNG

The stair-like characteristic of PV power curve is determined by the discrete updating step interval of the MPPT algorithm. PV power peak, which coincides with maximal irradiance level at 12 noon, should appear as expected.

5.3 System Power Flow Analysis

In the following figure, the power curves for the DC nano grid are illustrated over the entire 24-hour simulation, representing the power of PV, the battery power (positive when discharging, negative when charging) and the power flowing over the DC bus to the load.

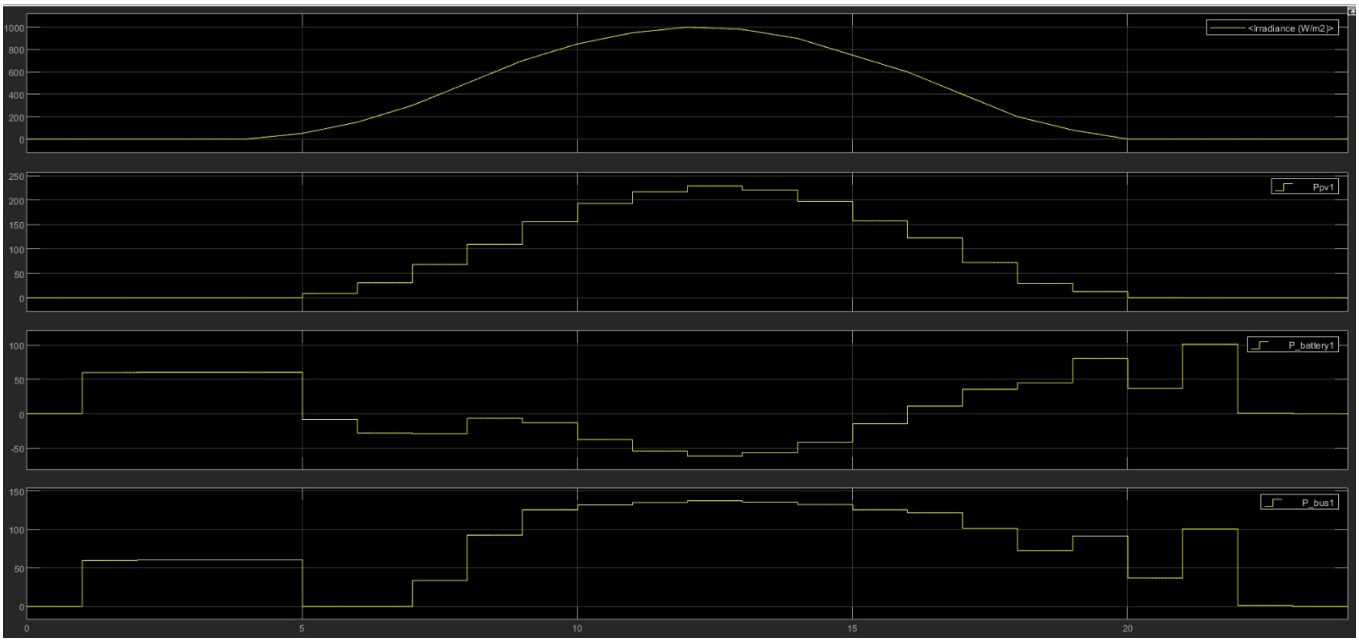


Figure 5.2: System Power Curves: PV, Battery and DC Bus over 24-hour period (Summer Irradiance)

During the daytime, the PV generation is greater than the load and so the excess generation will be used for charging the battery (negative battery power). During the morning hours (early morning) and evening hours (late evening), when the generation is less than the load, the battery discharges current (positive battery power). So, all three modes of battery control are correct.

5.4 Battery Performance Analysis

5.4.1 Battery Status: Charging=1, Discharging=-1, Idle=0

Fig. 5.3 shows battery operation mode status during the whole simulation, changing in charges (+1), discharging (1), and Idle state (0) according to PV generated power versus load required power balance.

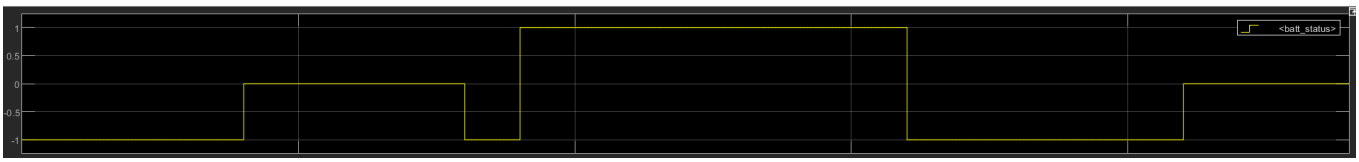


Figure 5.3 Battery status

5.4.2 Battery Voltage

The battery voltage stayed within safe operating limits throughout the simulation. It rose steadily during charging phases, indicating energy storage, and decreased in a controlled manner during discharge, showing stable energy delivery without sudden fluctuations.

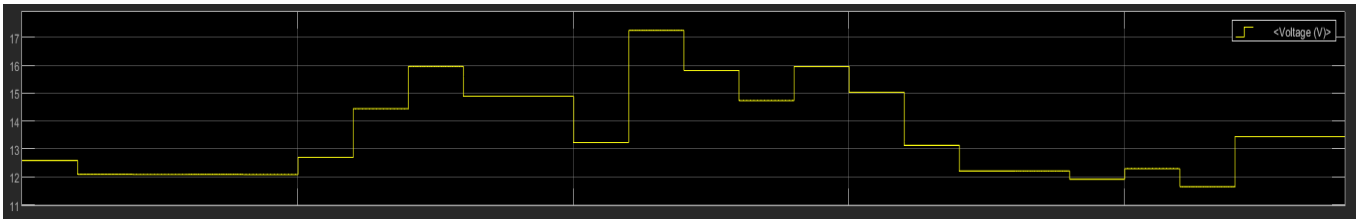


Figure 5.4 Battery Voltage

5.4.3 Battery Current

The direction of battery current clearly indicated system operation:

- Positive current refers to discharging (energy flowing out of the battery)
- Negative current refers to charging (energy flowing into the battery)

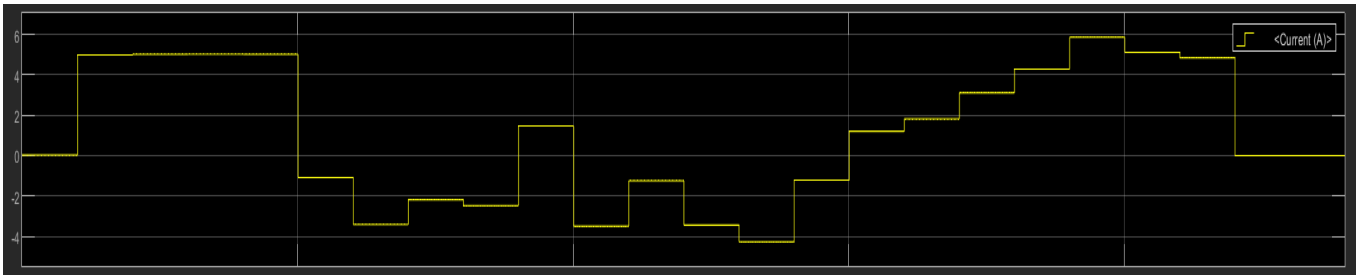


Figure 5.5 Battery Current

5.4.4 State of Charge (SoC)

The SoC profile effectively captured the system's energy behaviour:

- It decreased during load-supported discharge phases
- It increased under surplus generation conditions
- It stabilized near maximum charge levels

Notably, the cloud-based controller-maintained SoC within set thresholds, preventing over-discharge and enhancing battery lifespan.

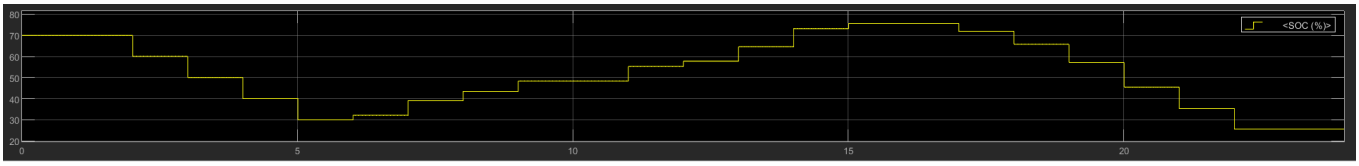


Figure 5.6 Battery SoC

6 Experimental Results and Hardware Validation

The final section introduces the outcome of hardware prototype verification. It shows the whole system, the PV panel, the MPPT controlled buck converter, bi-directional converter, the storage device and the sensing circuit, and dual-MCU control parts integrated and working in the real conditions. Three main tests were performed to evaluate the system function under each possible state of the operation.

6.1 System Integration and Test Setup

The following figure fully integrated hardware prototype during testing outside. All the subsystems are wired through the single 12V DC bus, and the dashboard was viewable through a live Firebase link during all the testings.

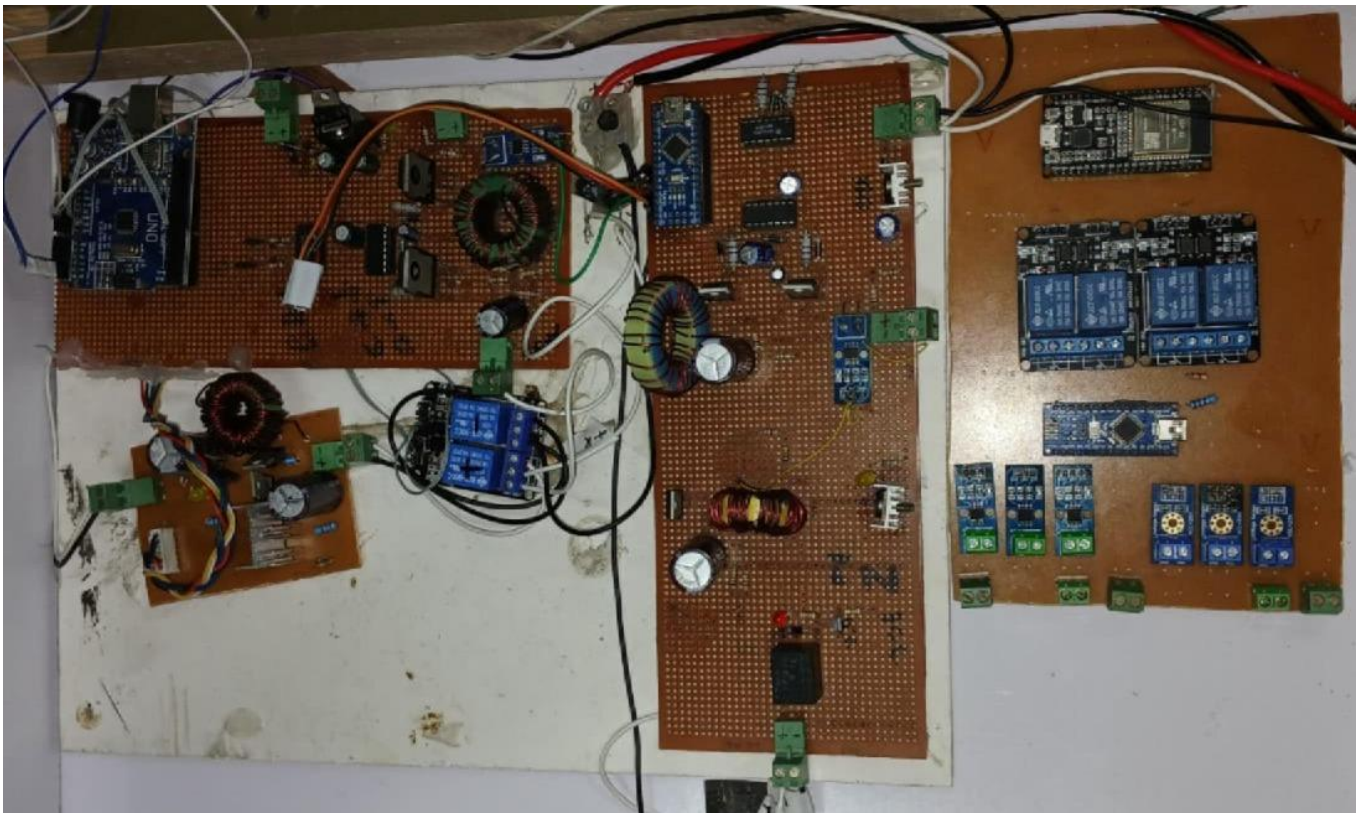


Figure 6.1 Complete fabricated circuit

The following figure shows the entire testing setup for a DCNG, comprising of a complete fabricated circuit, attached with a PV panel, and a battery and loads (Fan, LED bulb) connected to it:



Figure 6.2 Completer hardware testing setup

The sensing circuit was tested individually prior to integration testing. The bench integration of the sensing circuit is displayed in Figure 6.2 showing voltage and current was correctly measured for all channels prior to connection to the power stage.

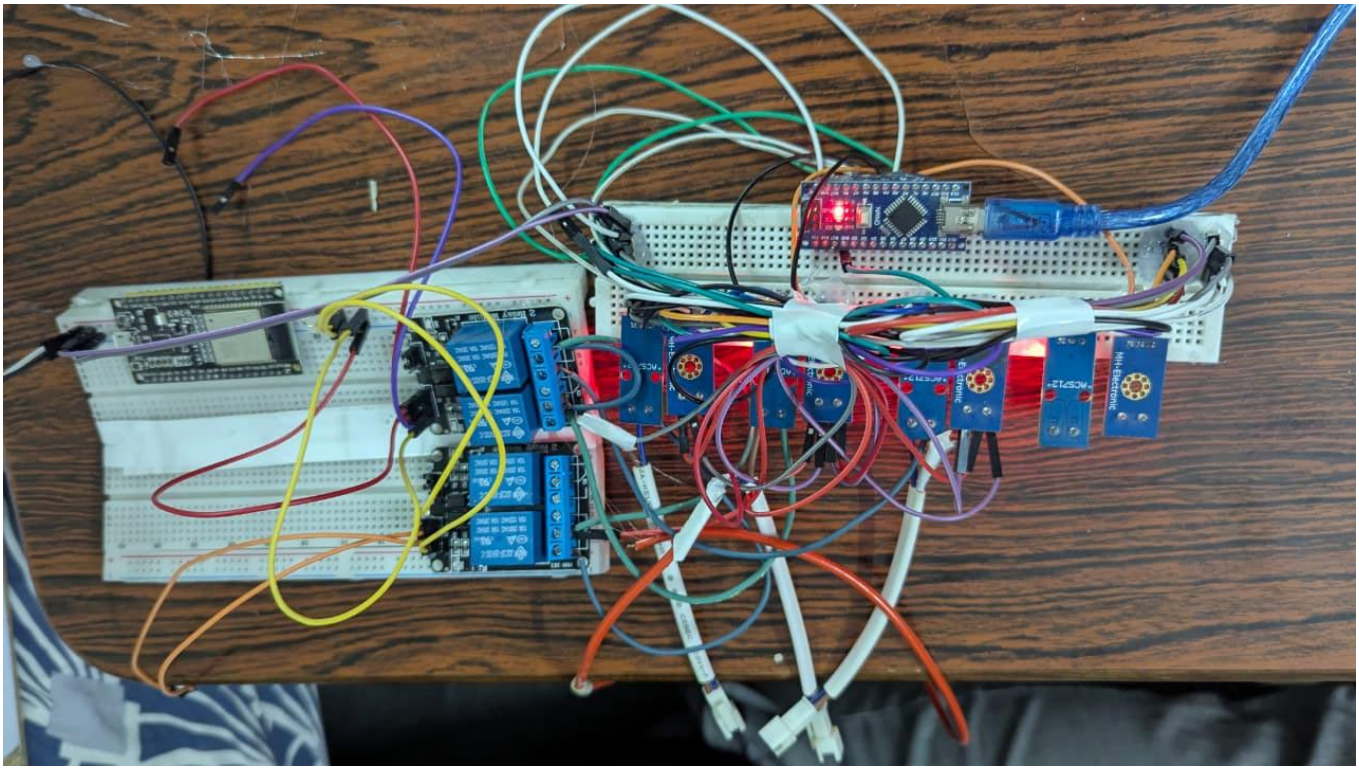


Figure 6.3 Sensing Circuit Integration

6.2 Testing Scenarios

6.2.1 Scenario 1: Battery Charging with No Load

This test demonstrated energy generation within each grid when solar power was available. A 180 W solar panel was used, and with no load connected, it charged the battery at 157.1 W output.

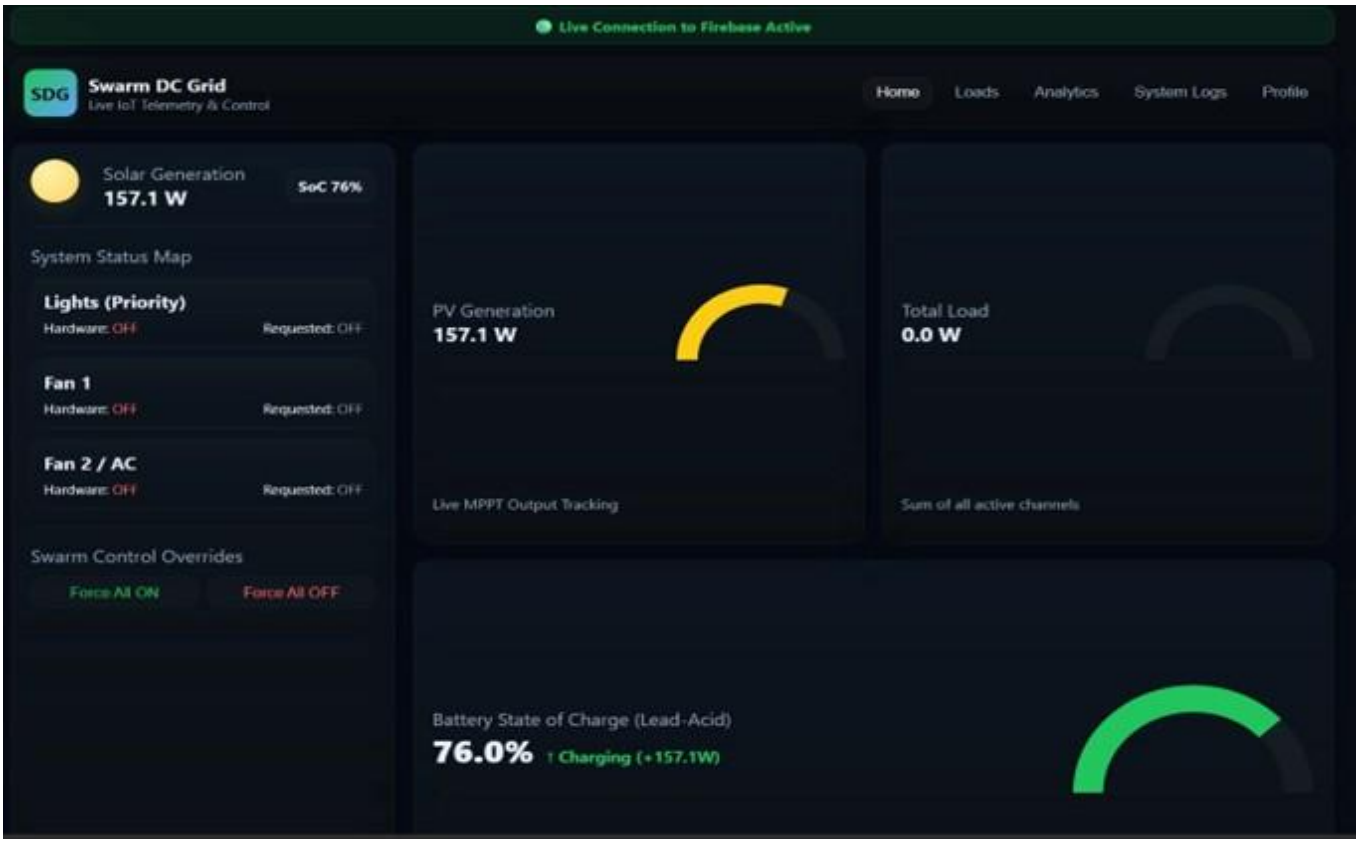


Figure 6.4 Dashboard – Battery Charging with No Load

6.2.2 Scenario 2: Battery Charging with Active Load

This test demonstrated energy generation within each grid when solar power was available. A 180 W solar panel was used which produced 83.2 W under cloudy weather, powering load of 50.8 W connected, and rest of the 32.4 W being used to charge the battery.

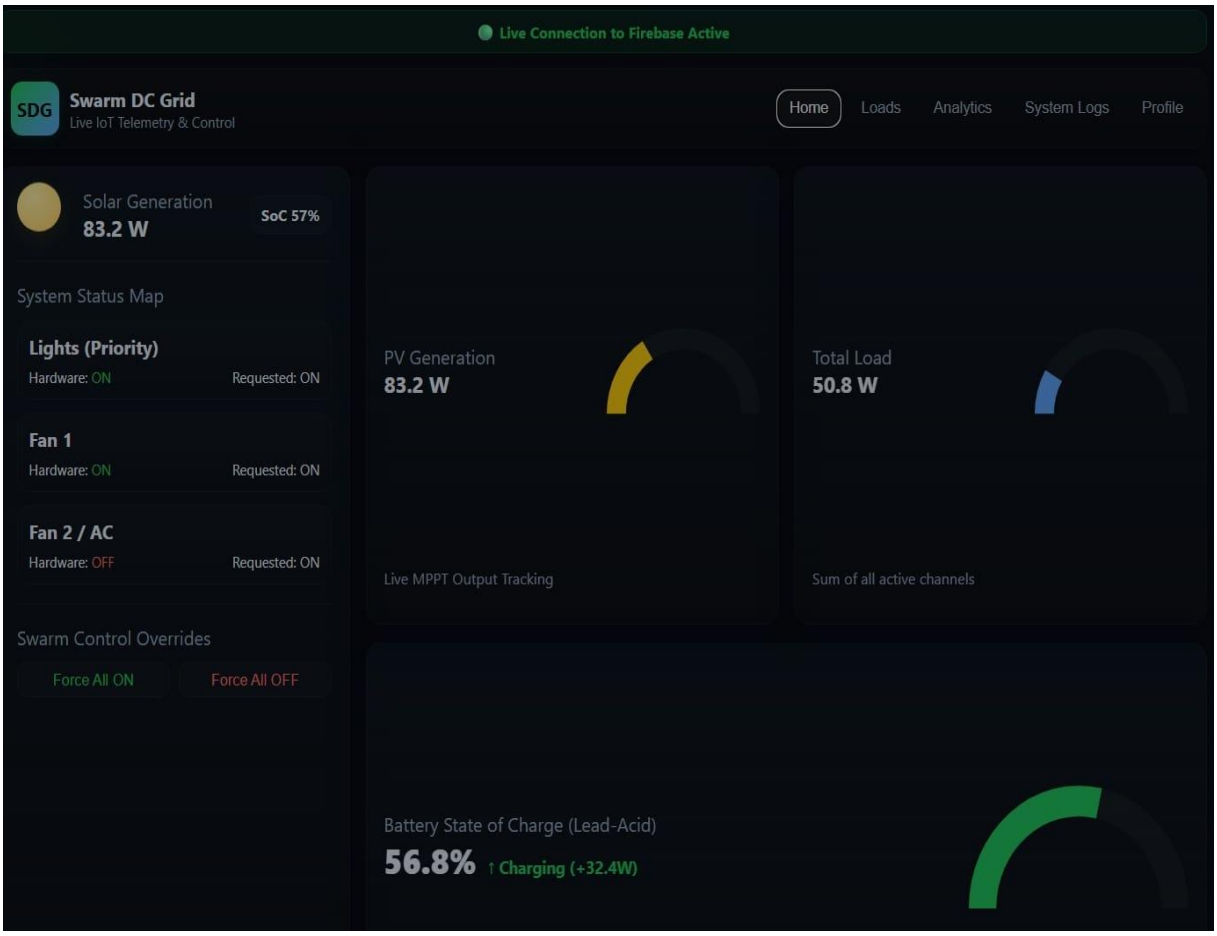


Figure 6.5 Dashboard: Battery Charging with loads connected

6.2.3 Scenario 3: Battery Discharging with No PV

This test demonstrated energy utilization within each grid when no solar power was available. A 12 V, 48 Ah battery is used to power the critical loads of lights and Fan 1, of 55.4 W while the SoC of the battery is shown to be 76%.

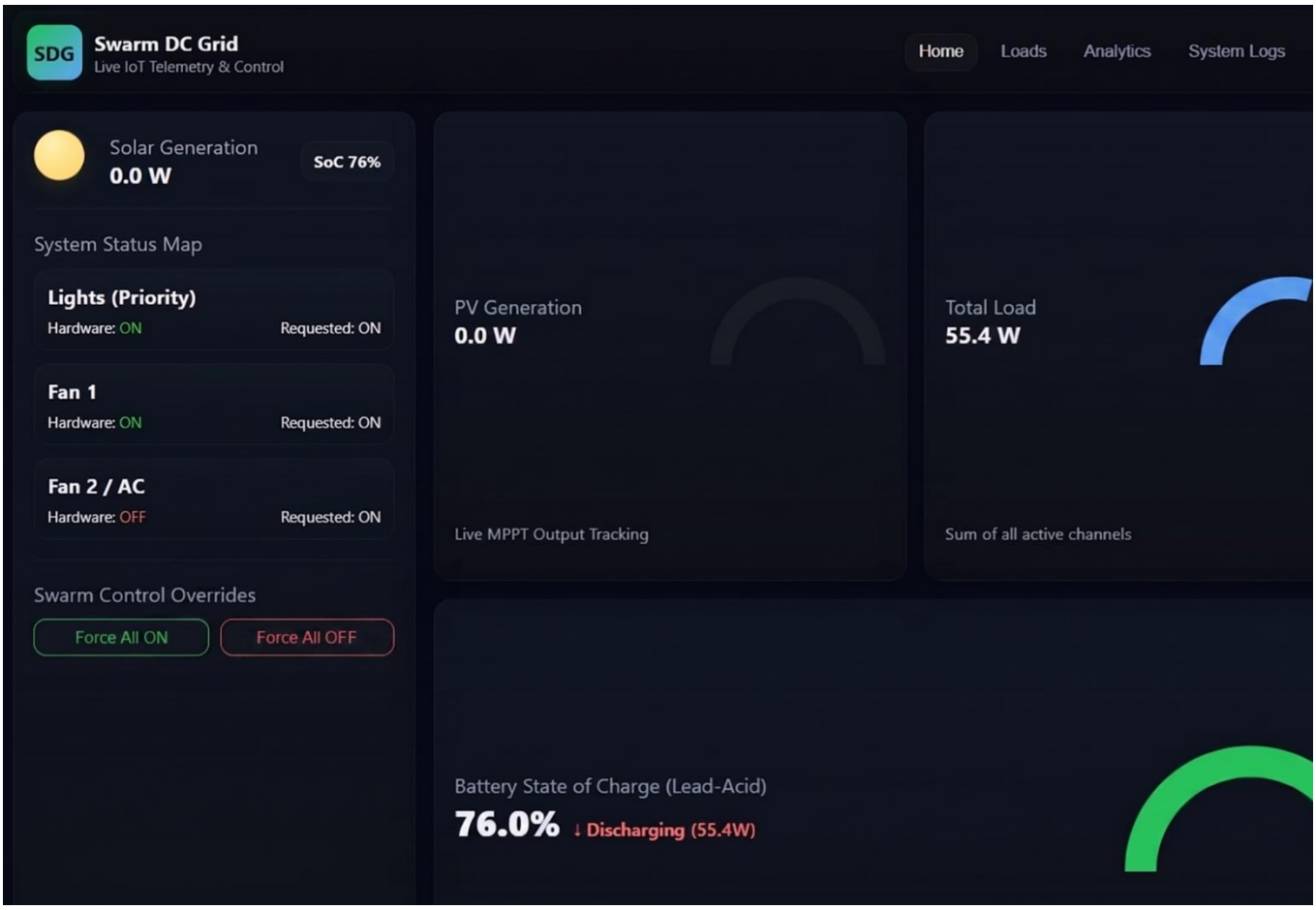


Figure. 6.6 Dashboard: Battery discharging with no PV

6.5 Hardware Challenges and Engineering Solutions

Table 6.1: Hardware Challenges and Resolutions

Issue	Root Cause	Solution
Erratic relay switching	Logic-level mismatch: ESP32 (3.3 V) driving 5 V relay module	Relay module powered separately from 5 V supply; logic level compatibility verified
Voltage drops under full relay load	Inadequate supply current for simultaneous relay switching (~70 mA each)	Upgraded to higher-current bench supply; improved wiring gauge
Damaged GPIO pin on ESP32	Sustained overcurrent during development	Remapped relay to an alternate GPIO pin; design flexibility preserved
UART interference during ESP32 upload	UART RX/TX pins shared with programming interface	Disconnected Nano–ESP32 UART link during firmware upload
Noisy sensor readings	Switching transients from buck and boost converters coupling into analog sensing lines	Applied 10-sample moving average filter in Arduino Nano firmware

6.6 Prototype Validation Summary

Hardware tests under each of the three scenarios demonstrated:

- Stable regulation of 12 V DC bus when varying PV and changing load
- Successful MPPT tracking, producing 87.3% power output of rating for the panel in full sun
- Appropriate changes of mode (charge, hold, discharge) for battery based on generation-load power
- Successful sensing of all sensor values in real time and consistent syncing with the Firebase cloud
- Appropriate working of the smart CEMS load optimization in terms of load priority and battery overrides in all test conditions

7 Conclusion

The design and implementation of the Swarm DC Nano Grid have been successfully demonstrated as a distributed architecture for decentralized rural electrification. The architecture combines Photovoltaic generation, energy storage in the form of a battery and Bidirectional AC-DC converters with IoT based control system integrated together.

Both simulation and hardware results show that DC bus voltage can be well regulated, the P&O MPPT algorithm is efficient in tracking maximum power and energy transfer between the nodes can be carried out reliably. Integrating with a cloud-based optimization layer using ILP enhances the performance of the Swarm system to prioritize loads, reduce the number of battery operations and ensures safety in case of different operating conditions. The feasibility of real time data sensing, communication and control were tested through the hardware prototype which also included implementation challenges like noise, communication constraints and power management issues.

This design is scalable, efficient in energy consumption and cost effective for rural off grid applications and further enhanceable by including other sources of energy as well.